A Survey on Traversability Analysis for Autonomous Legged Navigation

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***Abstract*—Traversability analysis asks whether terrain can be crossed safely and efficiently, not merely whether it is free of obstacles. Existing surveys answer that question for wheeled platforms. Their formulations assume continuous ground support, and the morphology they formalize is that of a vehicle. Legged robots do not share it. They move through discrete, intermittent contacts, so terrain suitability must be assessed at the level of individual footholds and of the body motion those footholds induce. This survey reviews traversability analysis from that perspective. It treats quadrupedal and humanoid platforms within a single analytical framework, and keeps them distinguished throughout. We adopt a robot-dependent definition of traversability, and organize the field along four axes: sensing, terrain representation, problem formulation, and estimation method. Sensing is grouped into exteroceptive, proprioceptive and multimodal families. Estimation is grouped into nine families, from geometry-based analysis to foundation-model and vision-language reasoning. Evaluation practice is reviewed separately, covering proxy ground truth, metric families, baselines and statistical strength. The comparison exposes a systematic asymmetry: quadrupedal work has moved from geometry toward semantic and physical interaction properties, while humanoid work has deepened the geometric representation and tightened its coupling to contact and balance, and neither has absorbed the other. We conclude that the clearest open direction is a representation carrying semantic and physical terrain properties, with calibrated uncertainty, at foothold resolution, and with the balance-aware conditioning that bipedal locomotion requires.**

I. Introduction

Traversability analysis is a central problem in field robotics, because autonomous systems must decide not only whether terrain is obstacle-free, but also whether it can be crossed safely and efficiently under the robot's physical constraints. Prior surveys define traversability as a function of both terrain properties and robot capabilities, contrasting terrain geometry and surface characteristics with the robot's kinematics and dynamics to generate a traversability or cost map for planning [1], [2], [3], [4]. The problem is hardest in unstructured environments, where irregular, deformable and partially occluded terrain, changing soil consistency, vegetation occlusions and hidden hazards require more than conventional obstacle avoidance [1].

Existing surveys are, however, organized around mobile robots, unmanned ground vehicles and planetary rovers, typically classifying methods into terrain classification, mapping or cost estimation, hybrid approaches and end-to-end learning, and further separating appearance-based, geometric and multimodal methods [1], [5], [6]. This wheeled inheritance is embedded in the formulation itself. Representative learning-based pipelines acknowledge that an estimate is conditioned on the robot's design and heading, and that a model calibrated for one configuration becomes obsolete once the dimensions or wheelbase change [7]. Morphology is therefore already a first-class variable in traversability estimation, yet the morphologies formalized so far are overwhelmingly those of continuously supported vehicles.

Legged robotics has meanwhile matured into a distinct research area centered on environmental perception, locomotion control, learning-based adaptation, sim-to-real transfer and deployment in unstructured environments [8], evolving from early hydraulic platforms such as BigDog and HyQ to lightweight electric quadrupeds like ANYmal, Spot and the Unitree and DEEP Robotics series that dominate current research [8]. Broader treatments surveying bipedal, quadrupedal and hexapod platforms together exist, focusing around gait planning, motion control, balance and actuation [9]. These reviews are broad in scope rather than dedicated syntheses of traversability analysis for legged systems. The gap matters because legged traversability is not a terrain-labeling problem: legged systems must reason about discrete footholds, contact transitions, gait and velocity adaptation, body stability and local interaction forces, which is why recent work relies increasingly on proprioceptive sensing, terrain probing, foothold-aware representations and adaptive coupling of vision-based and contact-based estimates.

The case for a legged-specific treatment strengthens once humanoids are included. Recent humanoid reviews organize the field around body structure, control and perception [10], establish taxonomies spanning perception, locomotion control, decision-making, foundation-model integration and human-robot interaction [11], and trace loco-manipulation from model-based contact planning to learning-based paradigms [12]. Crucially, this literature does not inherit the quadruped formulation of traversability. Instead, it reframes traversability around the foot-ground interface. Bipeds are supported by a small number of large flat contacts. As a result, environment representation becomes foothold-centric rather than cost-field-centric. This shift motivates dedicated work on several fronts: planar semantic maps [13], terrain segmentation that recovers stable six-degree-of-freedom footholds [14], and footstep planning performed continuously as a consequence of bipedal locomotion [15]. Revealingly, the open problems named by these reviews coincide with the core concerns of traversability estimation: how to estimate stiffness, damping, plasticity, heterogeneity and porosity, what metrics capture terrain complexity, and how to fuse tactile sensing with proprioception and vision [12]. Perception is independently listed among the

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principal bottlenecks, with humanoids still unable to walk stably on uneven heterogeneous surfaces [11], adaptive terrain recognition appears as a priority in optimization-based control [10] and identification of traversable regions is singled out for humanoid visual semantic understanding [13]. The humanoid community has thus repeatedly named traversability estimation as an open problem without consolidating it into a coherent body of methods.

A further limitation concerns how narrowly terrain has been defined. Prior reviews largely treat challenging terrain as stationary irregular surfaces. This leaves locomotion on accelerating substrates such as ships, trains and aircraft underrepresented. On such platforms, perception must additionally separate robot ego-motion and obstacle motion from exogenous platform motion [16]. The support surface is not always static, and formulations built on that assumption do not transfer without modification.

Different from existing surveys on general mobile-robot traversability, on broad quadruped robotics, or on humanoid perception and control, this paper reviews traversability analysis from the perspective of legged locomotion, treating quadrupedal and humanoid platforms within a single analytical framework. We structure the field around sensing and perception, terrain and foothold representations, traversability estimation methods and evaluation protocols, with particular attention to proprioceptive learning, multimodal fusion, adaptation to uncertain terrain and sim-to-real deployment. As Table I shows, surveys that place traversability at their center are written for wheeled platforms, while those covering legged morphologies treat it only in passing; this survey occupies the resulting gap.

TABLE I: Prior surveys related to this work. Traversability is the organising subject (●), one topic among many (◐), or not addressed (○). Morphology is quadrupedal (Q), bipedal (B) or hexapod (H). No prior survey combines a central treatment of traversability with legged morphology.

Survey	Yr	Scope	Trav.	Morph.
[5]	2020	Ground vehicles	◐	none
[4]	2022	Ground vehicles	●	none
[3]	2022	Mobile robots	●	none
[6]	2023	Ground vehicles	●	none
[14]	2023	Humanoid, percep.	◐	B
[2]	2024	Ground vehicles	●	none
[13]	2024	Humanoid, percep.	◐	B
[15]	2024	Humanoid, planning	◐	B
[10]	2024	Humanoid, general	◐	B
[9]	2024	Legged, general	○	Q, B, H
[1]	2025	Forestry, ground	●	none
[12]	2025	Humanoid, loco-man.	◐	B
[16]	2026	Legged, non-inert.	◐	Q, B
[11]	2026	Humanoid, general	◐	B
[8]	2026	Quadruped	◐	Q
This survey	2026	Legged traversability	●	Q, B

II. Scope and Definitions

This survey considers traversability estimation for legged robots operating in unstructured, natural or cluttered human-made environments, covering both quadrupedal and bipedal or humanoid platforms. We focus on methods that estimate, represent or exploit terrain traversability for perception, mapping, path planning and terrain-aware locomotion, treating

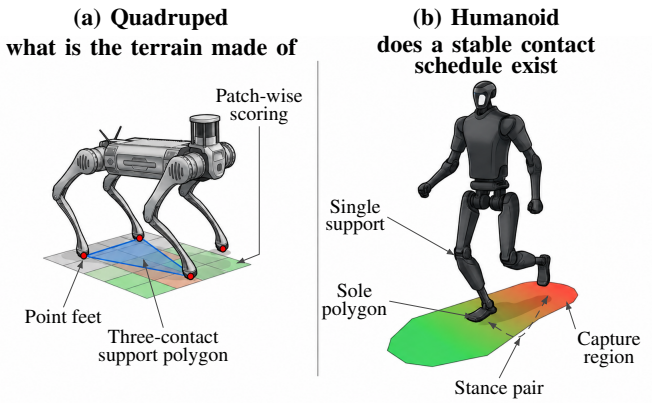


Fig. 1: Morphology sets the question. A quadruped stands on point feet, so its support region is the convex hull of the loaded contacts, three of them here, and terrain is scored patch by patch as a field of local attributes such as friction, roughness or stiffness. A humanoid spends much of the gait in single support on one sole polygon, so a patch is admissible only when a whole sole fits inside a planar region, candidate footholds are scored as stance pairs rather than as isolated cells, and the safe set is bounded by the capture region as much as by the terrain itself. The same ground therefore poses two different traversability questions.

traversability as an intermediate systems-level concept linking environment understanding to motion generation and control [17], [18], [19], [20], [21].

Quadrupeds and humanoids are treated jointly because they share the defining property that motion is realized through discrete, intermittent contacts rather than continuous ground support, so terrain suitability must be assessed at the level of individual contact locations and of the resulting body motion. They are nonetheless kept distinguished throughout, since morphological differences lead to different formulations, representations and failure modes, as summarized in Fig.1. The joint treatment is also motivated by the structure of the existing literature, in which humanoid surveys either address terrain perception as one component among many [13] or treat traversability only implicitly through contact and foothold planning [12], [15], while broader legged surveys concentrate on control and actuation [9].

The scope is broader than classical obstacle avoidance, since legged robots must reason about support quality, slip risk, foothold feasibility, terrain deformability and control difficulty [22], [23]. For humanoids it extends further, because the tall articulated body implies that traversability is not determined by the support surface alone: cluttered scenes may require crouching, hurdling or squeezing through narrow passages, so whole-body clearance becomes part of the assessment [24], and additional contacts such as palms on walls or railings change which regions are traversable at all [25], [26]. The scope remains narrower than the general legged robotics literature, since work on gait generation, balance recovery or whole-body stabilization is considered only when it constructs or consumes an explicit representation of terrain suitability, as in perceptive locomotion policies conditioned on elevation maps or foothold quality signals [27], [28], [29].

A. A robot-dependent definition

Traversability is generally defined as a robot's ability to move safely through an area, not an intrinsic terrain property, but one determined jointly by terrain characteristics and robot-specific factors such as embodiment, actuation, locomotion strategy, state estimation quality and control capability [30], [18]. This dependence is pronounced for quadrupeds, whose discrete footholds and adaptable gaits enable negotiation of terrain unsuitable for wheeled platforms while remaining sensitive to slip, entrapment and local support failure [31], [19].

For bipedal platforms the robot-dependent component becomes considerably more restrictive, for four reasons that recur throughout this survey. First, **coupling with dynamic balance**: with a single support foot during most of the gait cycle, admissible foot placements are constrained jointly by the terrain, by kinematic reach and by the capture region of the current center-of-mass state [32]. Second, **extended contact surfaces**: humanoid feet are polygonal rather than point-like, so foothold quality is properly expressed as the overlap between the sole and the safe terrain region, which invalidates point-foot assumptions inherited from quadruped formulations [28], [13]. Third, **locomotion mode**: traversability boundaries are calibrated to the current mode and may trigger transitions such as walking to crawling when the terrain no longer admits upright progression [33], the analogue of gait selection among trot, crawl and amble in quadrupeds [34], [35]. Fourth, **variable robot state**: the same terrain is not equally traversable at all times, since carried payloads alter the disturbance the controller must absorb [36] and degraded stability can justify restricting the perceptual horizon to priorities immediate foothold safety [37].

Traversability should therefore not be treated as strictly binary. Recent legged work increasingly models it as a continuous score or cost ranking terrain by expected safety, stability or energetic efficiency [31], [17], [18], which suits legged navigation, where multiple feasible paths may differ significantly in risk or locomotion cost. The bipedal literature illustrates the diversity of the resulting output spaces particularly well. At one end is a continuous mobility score regressed from slope and roughness [38]. Others predict the number of useful footstep combinations available for a torso transition [25], [26], or paired foothold and stance scores sampled around nominal foot placements [39], [20]. At the other end is a stability-aware command velocity, in which traversability acts as a constraint on the admissible action set rather than as a scalar cost, with uncertainty made explicit through a risk measure [21], in line with risk-aware quadruped work [40].

B. Three notions of traversability

This survey organizes legged traversability into three complementary notions. **Geometric traversability** denotes terrain suitability inferred from spatial structure, namely slope, roughness, step height, elevation variation and support-surface geometry, typically computed from depth or LiDAR. For quadrupeds this must account for embodiment, since obstacles that would block a wheeled robot may be passable through foothold adaptation or body motion [19], [18]. Geometric formulations dominate the humanoid literature and are frequently specialized to bipedal contact requirements, through elevation-and-normal grids [32], height-map sampling around nominal footholds [39], [20], index-based models evaluated against a robot-specific boundary [33], mobility maps [38], and hybrid planar-region and height-map representations [41], [42].

Semantic traversability denotes terrain suitability inferred from terrain category or scene interpretation, using labels such as sidewalk, grass, vegetation or low-friction surface. It is useful when geometry alone cannot distinguish safe from unsafe terrain, or when identical geometry corresponds to different physical conditions [43], [44], [45]. In the humanoid domain, traversable semantic maps have been identified as a key enabling technology [13], but concrete instances label geometric primitives such as staircase polygons rather than materials [42].

Physical or interaction-based traversability refers to terrain suitability inferred from the terrain's actual effect on mobility, through friction, stiffness, slippage, traction, energetic cost and stability margin. This notion is central because visually similar terrains may induce very different contact behavior. Active probing allows a legged robot to estimate friction and propagate that knowledge to visually similar regions [46], [47], while related work defines traversability through proprioceptively estimated slippage, energy cost and stability margins, or through learned prediction of friction and stiffness [48], [49]. In humanoid work this notion appears mainly through stability proxies learned from locomotion experience rather than explicit terrain parameters [21], [50], and therefore remains comparatively underdeveloped for bipeds.

These notions are complementary rather than mutually exclusive, and recent systems increasingly combine all three [51], [31], [19].

C. Granularity and representation

Orthogonally, legged traversability is expressed at two levels of granularity that this survey keeps distinct. At the body level it is attached to a body or torso pose and used to guide path search, as in torso-pose guiding paths whose edge costs encode learned estimates [26] or grid searches in which each edge models a single stride [39], [20]. At the contact level it is attached to candidate foot placements, as in reachable foothold extraction from planar terrain models [42], [41] and support-surface mapping feeding terrain-adaptive local planning [19]. The two are coupled, since body-level feasibility depends on the availability of admissible contacts, precisely the quantity regressed by predictors that count feasible footstep combinations [25]. The separation is sharper for bipeds, whose alternating stepping structure must be encoded explicitly in the cost formulation [20].

A further distinction concerns how the estimate is represented. In *explicit* formulations, traversability is materialized as a map or cost layer consumed by a separate planner [17], [39], [33]. In *implicit* formulations it is embedded in a learned model or policy and never exposed as a standalone map, as in forward dynamics models encoding traversability

through predicted failure risk [52] or perceptive humanoid policies conditioned on elevation maps or raw depth [27], [53], [54], [29]. Both are within scope, since in both cases terrain suitability is estimated and used to determine motion.

Accordingly, this survey adopts the following operational definition: legged traversability is the robot-dependent degree to which a terrain can support safe, stable and efficient legged motion, as inferred from geometric, semantic and physical interaction cues, and used for terrain-aware planning and control [30], [31]. For bipedal platforms this degree is additionally conditioned on the availability of admissible, kinematically reachable and dynamically recoverable contacts, and on whole-body clearance along the intended motion [32], [24].

III. Sensing for Traversability

Traversability estimation is ultimately bounded by what the platform can measure. Sensing choices cluster into three families: **exteroceptive, proprioceptive and multimodal**. These families capture complementary aspects of the problem: exteroception provides terrain awareness before contact, proprioception reveals how the terrain affects the robot during locomotion, and multimodal sensing couples both [19], [34], [55]. A fourth, degenerate family appears almost exclusively in **humanoid contact planning**, where the environment is supplied to the planner as a pre-computed set of contactable planar surfaces and no physical sensor is involved [25], [26]. These works isolate the planning contribution, but their estimates are never exposed to sensor noise, occlusion or registration error.

A. Exteroceptive sensing

Exteroceptive methods infer traversability from RGB, stereo, RGB-D and LiDAR measurements to obtain look-ahead information, detect obstacles, recover terrain geometry and build representations before contact [19], [56], [23].

The lightest configurations rely on a single color camera. Wild Visual Navigation [17] learns a dense traversability estimate from a wide field-of-view monocular stream alone, using the onboard LiDAR and depth camera only for local mapping and never for the traversability signal itself. Similar RGB-only pipelines drive semantic segmentation on computationally constrained platforms [57], exploit egocentric video from a chest-mounted pedestrian camera [43], reconstruct pose-projected features [45], and support map-free topological navigation in which monocular depth estimation replaces a depth sensor entirely [58], [59]. Stereo rigs extend this to metric reconstruction for terrain reconstruction and texture recognition [60] and, on the bipedal side, provide the point clouds from which elevation-map patches are built [21].

RGB-D sensing is the single most common exteroceptive configuration, since it delivers color and metric geometry from one compact device. Quadrupedal systems use it to recursively estimate terrain class and elevation [44], to derive gravity-aligned depth and surface-normal modalities for anomaly-based classification [56], and to feed interactive or risk-aware navigation policies [61], [62]. In the humanoid literature the dependence on depth is stronger still. Structured-light and stereo depth sensors produce the point clouds from which mobility maps are computed [38]. The same sensors incrementally update a stochastic elevation-and-normal grid [32], or supply low-resolution depth used directly as policy input [29]. They also stream depth for perceptive parkour [54]. In one payload-robust policy, depth is the only runtime exteroceptive channel, while the elevation map is reserved for the privileged critic during training [36]. Depth-only designs deliberately reduce the suite to a single camera, removing both LiDAR and external localization [53], and depth also forms the front-end of conventional bipedal navigation stacks [63], [64].

LiDAR is preferred where long range, wide field of view or illumination invariance matter. Purely geometric traversability has been estimated from 3D LiDAR alone through self-supervised anomaly detection [51], and LiDAR scans projected into occupancy grids at several heights allow vegetation to be separated from solid obstacles [65]. In humanoid path planning, high-resolution LiDAR is the standard mapping sensor, building the 2.5D height maps [66] over which foothold and stance traversability are assessed [39], [20], scanning the environment for geometric terrain indices [33], and supporting torso-mounted odometry [63]. Other systems rely on LiDAR-inertial SLAM and an occupancy representation rather than raw images, arguing that the resulting potential field acts as a low-pass perceptual filter [24], or operate on organised point clouds without committing to a specific sensor [41].

Despite its predictive strength, exteroception degrades under occlusion, poor illumination, dense vegetation or low-feature environments, and cannot capture latent physical properties such as deformability, softness or slip risk [34]. Low-friction surfaces such as ice are poorly represented by geometry-based maps [44], penetrable vegetation causes raw LiDAR returns to sit well above the true support surface [67], and structured-light sensors suffer from daylight interference [38].

B. Proprioception sensing

Proprioceptive methods infer traversability from the robot's internal state and contact response, using IMU measurements, joint encoders, torques, force or current sensing, contact states, slippage and stability or energetic indicators. These signals are especially informative for legged machines because the legs physically probe the terrain during locomotion [31], [34], [48].

The minimal configuration is an IMU combined with joint encoders. Vibration, angular velocity and foot-body motion relationships characterize roughness, slope, softness and slippage, and this pairing alone supports multi-label terrain classification when supplemented only by the control commands [31]. The same class of signals extends from classification to mapping, with joint encoders, an IMU and foot contact sensors, together with joint torques, velocities and base accelerations, aggregated into a multi-layer 2.5D grid encoding elevation alongside robot-centric interaction metrics, with no exteroceptive sensor at all [48].

Contact-level sensing gives direct access to the mechanical response of the terrain. Six-axis force-torque sensors on each foot produce the ground reaction score used to label images for

self-supervised property prediction [68], and joint encoders, force sensors and battery current sensors are combined into a proprioceptive vector from which terrain stability, motion resistance, entrapment risk and crash risk are inferred [34]. Active variants deliberately generate contact information rather than waiting for it, measuring terrain collapsibility with a force sensor on a probing arm [22] or exciting the proprioceptive channels through pressing and rubbing motions from which friction is regressed [46].

The main limitation is that proprioception is largely reactive: signals gathered during or after contact offer a faithful robot-centric measure of terrain already being traversed but no look-ahead [34]. This is compounded on soft terrain, where slippage corrupts leg odometry and hence state estimation even when the terrain appears visually benign [18].

C. Multimodal sensing

Multimodal methods combine exteroceptive look-ahead with proprioceptive confirmation or correction, and represent the dominant design pattern in recent legged traversability systems [19], [55], [23].

Support-terrain mapping fuses LiDAR point clouds, depth images and IMU-based localization with foot contact forces, linking terrain reconstruction to contact-aware analysis [19]. TOP-Nav assigns RGB to terrain classification, depth to obstacle detection and proprioception to an advisor that corrects the vision-based cost map on novel terrain [55], and traversability-aware navigation similarly combines color, depth and proprioceptive feedback to estimate cost in a robot-centric way [23]. AMCO [35] extends the idea to sensor reliability, weighting a stereo and LiDAR semantic map against proprioceptive deformability estimates according to measured image brightness and motion blur, an explicit reliability estimate that GrASPE [69] also computes over RGB-D, LiDAR and odometry before graph-based fusion. Richer suites appear in state estimation and physical-property inference, fusing IMU, LiDAR, depth, torque sensors and joint encoders to estimate state and terrain stiffness on deformable ground [18], pairing RGB-D and LiDAR with joint states and contact forces for online friction mapping [46], and aligning egocentric RGB, IMU, proprioceptive, odometry and tactile channels in a shared latent space [70]. LiDAR and RGB are further combined for support-surface recovery under vegetation [67], continual online learning [71], [30], risk-aware off-road planning [40], adaptation supervised by proprioceptive roughness [72], and traversability-aware Gaussian-splatting maps [73]. Height scans derived from depth cameras are combined with base velocities, projected gravity and joint states in learned forward-dynamics models [52], with a 2D line-scan LiDAR playing the same role in an earlier variant [74], while vision-language stacks add RGB, depth, IMU and stereo visual-inertial odometry on top of proprioception [75]. Cross-modal self-supervision links the two families explicitly, with a proprioceptive and geometric decoder supervising a visual network that predicts friction and stiffness per pixel [49], and Bayesian fusion of RGB-D with a tactile probe correcting semantic misclassifications that would otherwise cause slipping [47].

Bipedal systems adopt the same pattern with a characteristic composition: an exteroceptive channel dedicated to elevation mapping, IMU-based LiDAR-inertial odometry for localization, and joint-level proprioception feeding the locomotion policy. BeamDojo [28] fuses LiDAR with IMU data to build a robot-centric elevation map, explicitly preferring LiDAR over depth cameras because the latter restrict motion to the forward direction. The Perceptive Internal Model validates two interchangeable configurations for the same elevation map, either a single LiDAR providing both odometry and point clouds or a depth camera paired with a tracking camera [27], and the Humanoid Perception Controller likewise derives point clouds and odometry from onboard ranging sensors [50]. Combined LiDAR and depth suites are used where fields of view must be complementary, supporting polygonal staircase mapping [42], a fused bird's-eye-view (BEV) representation [37] and map-based localization [29]. Depth-only humanoid configurations still pair the camera with proprioception, either to reconstruct local heightmaps [53] or to drive an end-to-end parkour policy [54].

D. Sensor suites across morphologies

First, *depth and LiDAR dominate humanoid work while color is comparatively marginal*. Almost every bipedal method builds an elevation map, a height map or a set of planar regions, and several use no color information whatsoever [39], [20], [28], [53], [24], [32], [41]. This follows from the morphology, since a biped has a small support polygon and one foot in contact for most of the gait cycle, so foothold-level geometry is the binding constraint, whereas semantic appearance, which dominates quadrupedal off-road work [65], [70], [58], is secondary.

Second, *there is no proprioception-only traversability estimation for bipeds*. Proprioceptive-only methods form an established quadrupedal category [31], [48], [34], but every surveyed bipedal method that uses proprioception uses it jointly with an exteroceptive elevation map, and largely as locomotion-policy state rather than as a terrain descriptor. Where proprioceptive information does define traversability on a biped, it enters only as a self-supervision signal at training time [21].

Third, *contact and force sensing is nearly absent on humanoids*. Foot force-torque sensing [68], [60], contact forces [19], [46], current sensing [34], tactile probes [47] and dedicated probing arms [22] are all represented in the quadrupedal literature. The bipedal works are far sparser. Only iWalker carries a contact channel, in the form of binary foot switches [64]. TACT goes the other way, explicitly avoiding force sensing and instead absorbing payload-induced wrenches through a virtual-wrench training scheme [36]. As a consequence, terrain physical properties such as friction, stiffness and collapsibility, routinely estimated on quadrupeds [49], [18], [22], [44], remain essentially unaddressed for humanoids, despite bipedal balance being considerably more sensitive to them.

IV. Traversability Problem Formulation

Traversability estimation for legged robots assigns a measure of expected navigability to terrain regions, foothold

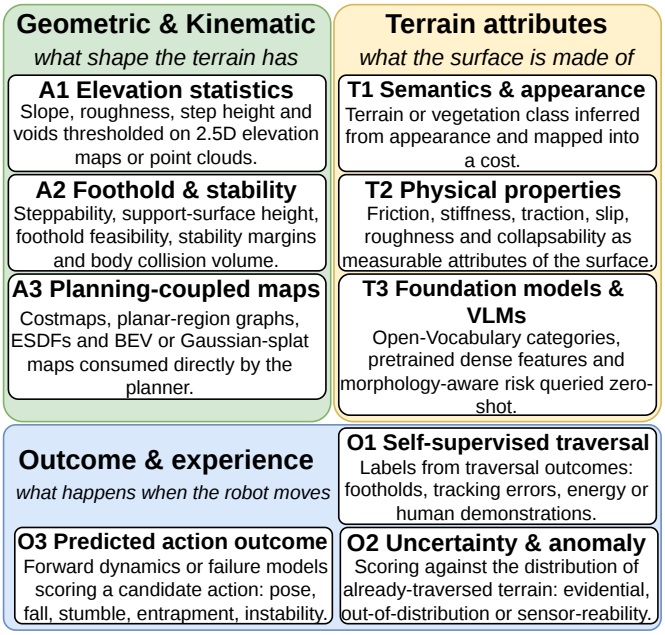


Fig. 2: Taxonomy of the cues used to estimate traversability for legged robots, grouped into geometric and kinematic descriptors (A1–A3), terrain attributes (T1–T3) and outcome or experience signals (O1–O3). Most systems combine cues from more than one family.

candidates or motion trajectories. This measure depends on the robot embodiment and on the locomotion controller. Conventional mobile-robot traversability is dominated by terrain geometry and collision avoidance; legged traversability must also account for discrete footholds, body motion, terrain contact, gait adaptation and stability [19], consistent with a literature that treats perception, sensing and motion control as coupled components [6]. Given sensory observations, the system estimates a representation that supports downstream decisions, which may take the form of a binary label, a continuous cost, a spatial or physical-property map, or a trajectory-level success probability [23], [22], [52].

The cues from which this representation is derived can be grouped into three families, summarized in Figure 2: geometric and kinematic cues describe the shape of the terrain and its compatibility with the robot's contacts, terrain-attribute cues describe what the surface is made of, and outcome and experience cues describe what happens when the robot actually moves over it. The formulations discussed below combine these families in different proportions.

A. What the estimate outputs

For quadrupeds the output is not used only to avoid obstacles. It also serves to select safer paths, adjust gait, regulate velocity and choose footholds [19]. The bipedal literature adds two output types that are rare in the quadruped case. The first is a *velocity-limit* output, in which traversability is the fastest velocity that keeps predicted instability below a threshold, passed to the planner as a constraint rather than as a cost [21]. The second is a *locomotion-mode* output, in which the estimate selects between qualitatively different body configurations [33]. Both reflect the same fact: for a biped, the safe response to difficult terrain is more often to change how the robot moves than where it goes.

Whether traversability is a binary feasibility problem or a graded cost was addressed in Section II-A. That distinction plays out clearly here. Early geometry-oriented classification by slope, roughness or obstacle structure [22] suits map-based planning. But physically passable terrain may still differ in energy cost, slip risk and controller difficulty [23]. Recent systems therefore rank candidate regions or trajectories by expected locomotion quality rather than by mere admissibility [48].

In the bipedal literature the graded formulation is anchored to foot and stance geometry instead of to a generic cost. Its recurring structural feature is that admissibility is aggregated over the contact area the robot actually requires, not evaluated at a single point: methods sample the height map around nominal foothold positions at a fixed stance width [43], [20], measure the overlap between sampled sole points and safe terrain [32], erode extracted planar regions with the foot footprint [42], or penalize placements near detected terrain edges using foot volume points [58]. This area-wise treatment is almost a necessity for humanoids: the foot is large relative to the terrain features that matter, and a partially supported foot produces an unrecoverable ankle torque. Quadrupedal foothold scoring, by contrast, can remain point-wise [70].

B. Formulation families

Beyond the geometric, semantic and physical notions introduced above, three further formulations organize the literature. A **controller-aware** formulation treats traversability as a function of the robot's locomotion capabilities rather than as an absolute terrain property. It explicitly critiques human-labeled environmental features as insufficient [23]. Multimodal variants combine color, depth and proprioceptive feedback, implicitly encoding terrain physics and controller stability [35]. Risk-aware variants plan against distributional traction instead of expected traction [40]. Perceptive humanoid locomotion takes this idea to its limit: in a large family of end-to-end reinforcement-learning (RL) controllers, traversability is never materialized as a map but remains implicit in the policy's value function and reward structure [27], [50], [53], [54], with foothold admissibility embedded directly in the training loop [29].

A **contact-configuration** formulation is essentially specific to humanoids, defining traversability over the contact configuration space rather than over the terrain surface. A humanoid can recruit other limbs in addition to feet, so the number of admissible contact combinations at a given torso pose varies enormously, and this variation becomes the quantity worth predicting. Traversability is then defined as the number of useful footstep combinations for a torso transition and used as a search heuristic biasing a contact-space planner away from regions with poor contact options [25]. Regressed per motion mode, the same quantity also segments a guiding path so that difficult segments are delegated to a more expensive planner [26]. Traversability here is a proxy for planner tractability rather than for physical risk, an interpretation with no counterpart in the quadruped literature.

A **whole-body clearance** formulation requires that the entire body clears the environment, not only the feet. Hierarchical bipedal planners separate the two concerns explicitly, using planar regions for footstep placement and body-collision avoidance while reserving a heightmap for swing-foot clearance, and distinguishing traversability from the cross ability of the surrounding structure [41]. Whole-body formulations replace the scalar cost with a differentiable directional field sampled at many body links [24], and index-based pipelines reach the same conclusion by allowing a change of locomotion mode when the space ahead is too narrow to pass upright [33]. Together, these works generalize body traversability [19] from a footprint-plus-clearance test to a full articulated-body admissibility problem.

C. Prediction level and planning interface

The formulation also depends on the level at which traversability is predicted.

- **Map level.** The output is a spatial representation of terrain quality.
- **Foothold level.** The output indicates whether a specific contact point offers adequate support [68].
- **Trajectory level.** The output evaluates whether a proposed path is likely to succeed under partially observed terrain [52].
- **Stride level.** This intermediate level is introduced by bipedal systems. A graph edge models a single step. The score is the availability of a paired left and right foothold configuration at the required stance width. The alternation structure of the gait is therefore encoded in the search itself [39], [20].
- **Footstep level.** Costs may be evaluated continuously, by sampling a signed-distance field at predicted footstep locations [64]. They may also be evaluated discretely, by emitting eroded foothold candidates [42].
- **Base level.** At the opposite extreme the terrain is flat and the dominant difficulty is dynamic obstacle avoidance. Bipedal navigation is then deliberately reduced to the wheeled case, modelling the robot as a differential-drive base over layered 2D costmaps [63]. The validity boundary of this reduction is itself a useful way to characterize when legged-specific reasoning is required at all.

The planning interface varies as well, from cell-wise costs for a planner, to terrain-aware high-level commands such as velocity, gait type and foot height [19], to hierarchical systems integrating terrain, obstacle and proprioceptive estimation within a combined planning and control framework [55]. This range carries a trade-off worth stating explicitly: explicit traversability maps remain interpretable, auditable and reusable across controllers and platforms, but must approximate the controller they serve, whereas implicit, policy-embedded traversability is exact for the controller that produced it but is not transferable, cannot be inspected and cannot be composed with an external planner. Several recent humanoid systems occupy a deliberate middle ground. One exposes a clean SE(2) velocity interface, so that a conventional planner can be layered on a perceptive controller [29]. Another gates distal perceptual information as stability degrades. The spatial horizon over which traversability is evaluated then becomes a function of the robot's dynamic state [37]. Hierarchical quadruped systems reach a similar compromise from the opposite direction, feeding controller-derived motion evaluations back into an explicit cost map [55].

V. Terrain Representation and Traversability Mapping

Terrain representation defines how sensory observations become a spatial model supporting locomotion and planning. Recent quadruped methods encode more than obstacle geometry, adding support quality [19], [67], local terrain cost [35], [55], [23], semantic category [17], [45], [65] and interaction-dependent physical properties [49], [44], [46]. Extending the analysis to bipedal platforms shows that this is not a platform-neutral design choice: the representation mirrors the contact and balance model of the robot that consumes it. A quadruped tolerates surface properties averaged over the footprint. A biped must resolve individual foot-sized patches, express the pairing between consecutive footholds, and account for a tall narrow body that may collide with the environment well above the ground plane. The discussion below groups both bodies of literature by the spatial abstraction they construct. Table II summarizes the comparison.

Geometry-centered elevation and occupancy maps. A common starting point is the geometry-centered map built from depth or LiDAR data, such as elevation, slope, roughness and occupancy-like grids, valued for spatial explicitness and direct use by conventional planners. Terrain probing combines elevation, roughness, slope and segmented vegetation cues into a grid [22], and support-terrain mapping adds contact-aware updates to build a real-time 2.5D [66], [76] support-surface representation [19]. The same grid dominates humanoid work, but its role differs: in quadruped pipelines it is one layer among several, consumed by an explicit planner, while in perceptive humanoid locomotion it is frequently the entire exteroceptive state, a small robot-centric patch fed directly to a learned policy [27], [28], [36], [29], [50]. Classical humanoid planners share the substrate while retaining an explicit search stage [39], [20], [21]. It is the single representation that both communities share without modification.

Foot-geometry-aware and contact-surface abstractions. The two literatures separate first on the fidelity demanded of that geometry. Quadruped formulations largely inherit a point-foot abstraction, adequate for small circular feet and dense foothold choice. Bipeds have large polygonal soles and few contacts, so the terrain model must effectively be convolved with the foot shape to become actionable [28], [54], [42]. This motivates a family essentially absent from the quadruped literature: explicit planar-region and contact-surface primitives. Feasible planar regions handle footstep selection and body-collision exclusion, with a heightmap for swing-foot clearance [41]. Others fit polygonal staircase maps by edge extraction and plane fitting [42], fit surface normals around nominal foothold positions [39], [20], or model the

environment as sets of contactable planar surfaces, including those usable by palm contacts [25], [26]. A non-grid variant departs from planar regions altogether [33]. It first builds an alpha-shape mesh of the terrain projected ahead of the robot, giving a continuous surface rather than a discretized grid. The robot's own body is then represented by an adaptive ellipsoidal model. Traversability follows from intersecting the two: where the body model clears the terrain mesh, the region is admissible. Segment-level abstraction is the one point of convergence: humanoid mobility maps score region-growing segments obtained from RGB-D normals [38], structurally the same move as the superpixel and segment-wise costs of vision-based quadruped pipelines [45], [17], [43], except that the segments derive from geometry rather than appearance.

Multi-layer maps and uncertainty. Multi-layer maps store several terrain attributes per cell: the soft-terrain quadruped framework stores inclination, roughness, foothold plane and stiffness [18], while humanoid multi-layer cells instead describe geometric confidence, as in the stochastic grid holding elevation, surface normal and measurement standard deviation [32]. Both communities have moved from deterministic to distributional cells, but quadruped representations place the distribution over terrain properties such as friction or traction [44], [40], and humanoid ones place it over the geometry itself, because for a biped the dominant risk is mis-stepping on a mis-measured surface rather than slipping on a correctly measured one.

Semantic and physical-property maps. A parallel direction assigns cost from terrain class or appearance rather than geometry [45]. Self-supervised approaches project robot interaction data into image space to build maps of learned terrain difficulty [68], [17]. Related work filters measurements semantically to recover the support surface beneath penetrable vegetation [67], or classifies vegetation by pliability before writing it into the cost map [65]. This axis has essentially no humanoid counterpart. Humanoid papers use the word semantic, it denotes a geometric primitive class such as a stair tread rather than a material category [42]. Terrain-property maps go further and encode physical quantities directly: a property-aware representation can identify low-friction icy surfaces that conventional maps fail to characterize [44], and such estimates are refined through tactile-visual Bayesian fusion [47], predicted from vision [49], annotated by active probing [46], or measured as collapsibility [22]. No surveyed humanoid representation stores a physical terrain property. Bipedal controllers are evaluated on low-friction and deformable surfaces [50] and compliance is handled at the policy level [36], but the property is absorbed into the controller rather than written into the map. Given that a biped loses balance on a low-friction patch far more readily, this is arguably the widest open gap between the two literatures.

Proprioceptive and experience-based maps. Proprioceptive maps encode how terrain feels to traverse rather than what it seems. Internal sensing feeds a multi-layer 2.5D gridmap of elevation, foot slippage, cost of transport and stability margins [48], supporting path comparison based on effort and mobility risk. Related frameworks reduce proprioceptive signals to a low-dimensional stability descriptor [34], or fuse the proprioceptive map with vision-derived expectations by reliability weighting [35], [55]. Bipedal work keeps the self-supervised principle but changes the stored quantity, supervising an elevation-map-indexed predictor with an instability signal selected for its correlation with fall risk [21]. The shift from slippage and cost of transport to fall probability reflects a difference in failure semantics: poor terrain degrades quadruped performance, but for a biped it terminates the episode.

Embodiment- and action-conditioned representations. A further trend dissolves traversability as a scalar field attached to the ground, replacing it with a function of the robot's state and intended action. Quadruped work scores sampled action sequences with learned forward dynamics by predicted pose and failure risk [52], [74], models distributions over achievable traction conditioned on commanded velocity [40], or uses a controller value function as the terrain cost itself [23]. Humanoid work reaches the same conclusion from the contact side: traversability becomes the number of useful footstep combinations for a torso transition [25], a per-mode regression over an SE(2) grid [26], a split into foothold and stance scores [39], [20], the fastest command velocity keeping predicted instability below a threshold [21], or the intersection of the traversable set, the kinematically reachable set and the one-step capture region [32]. Both communities therefore condition traversability on embodiment. The conditioning variable is a continuous velocity command for quadrupeds and a discrete contact configuration for bipeds.

CAT adds a third way to condition on embodiment [77]. The conditioning variable is neither a velocity command nor a contact configuration. It is a capability profile that a vision-language model derives from a language description of the robot. The profile down-weights terrain classes the platform cannot use, so one model can serve several embodiments. This directly targets the concern that a model calibrated for one configuration becomes obsolete when the platform changes [7].

Beyond the ground plane: volumetric and continuous fields. Because a humanoid is tall and narrow, several representations abandon the ground surface entirely. One encodes the humanoid-obstacle relationship as a continuous differentiable gradient field, supplying each body segment with a direction of safe motion [24]. Another builds a Euclidean signed distance field from projected depth, so that collision costs remain differentiable at sampled footstep locations [64]. A third excludes planar regions whose occupancy would place the torso in collision [41]. A fourth populates layered 2D costmaps from plane-segmented point clouds [63].

The quadruped literature reaches comparable constructions from dense clutter and pliable vegetation: safe corridors built from ellipsoidal Gaussian primitives [59], a traversability-aware Gaussian splatting map converted into a signed distance field fused with a LiDAR one [73], and occupancy sliced at multiple heights to separate tall rigid obstacles from traversable grass [65]. The convergence on continuous fields is genuine, but it is driven by whole-body collision on humanoids and by penetrable clutter on quadrupeds.

Implicit and learned latent representations. Finally, a

growing number of methods replace or augment the explicit map with a learned latent space. Quadruped approaches embed terrain appearance with self-supervised features and score novelty by reconstruction error [45], [17], align vision with inertial, proprioceptive and tactile signals [70], cluster autoencoder embeddings of traversed patches [30], [71], or construct a positive hypersphere in a BEV latent space [51], [56]. Humanoid latent representations instead reconstruct or attend to geometry: local heightmaps are reconstructed from egocentric depth, making the elevation map a learned output rather than a sensed input [53]. LiDAR and depth are fused into a BEV feature volume with a stability-aware gate that truncates distal information when balance degrades [37]. End-to-end parkour maps raw depth directly to joint actions with no intermediate map at all [54]. The gated variant is notable because it makes the effective spatial extent of the representation a function of the robot's dynamic state, an idea with no direct quadruped analogue.

Spatial extent, resolution and planning interface. Representations also differ in spatial extent and interface. Local maps support short-horizon path generation and immediate locomotion adaptation, global maps longer-horizon route selection [18]. Humanoid systems commonly pair a coarse global costmap with a finer local map refreshed at higher rate [63], or search a global height map before refining the path by continuous optimization [39], [20]. Resolution requirements differ in kind, bipedal local maps must approach sub-foot resolution to be actionable, while quadruped local maps can remain coarser without losing foothold feasibility, and the interface varies from dense costmaps to support-surface maps for gait and foot-height adaptation and to sets of contact primitives for footstep search [41], [26]. Table II summarizes these formulations side by side. Its final row states the open direction that their complementarity identifies.

VI. Traversability Estimation Methods

We organize existing methods along two orthogonal axes. The primary axis is the source of the traversability signal and the learning paradigm used to extract it, giving nine families: (i) geometry-based, (ii) foothold-, stance- and stability-aware, (iii) semantic and appearance-based, (iv) proprioceptive and terrain-property estimation, (v) self-supervised and experience-based, (vi) uncertainty-aware and anomaly-detection, (vii) predictive and policy-implicit, (viii) planning-coupled and full-stack systems, and (ix) foundation-model and vision-language approaches. The second axis is morphological, and it is not merely a matter of hardware: a statically stable quadruped can treat traversability as a property of a terrain patch, since three contacts remain to absorb a misjudgement, whereas a biped must reason about whether an admissible sequence of contacts exists and whether the gait stays dynamically viable between them (Fig. 1). The families are not mutually exclusive, and the strongest recent systems are hybrids.

A. Geometry-Based Methods

The classical approach infers traversability from 3D structure, typically 2.5D elevation maps [66] yielding slope, roughness, step height and foothold quality. Geometry remains the backbone of most modern pipelines, as the first layer of a hierarchical framework [22] or as dense support-terrain maps fusing LiDAR, depth and foot contacts [19]. GSAT shows that geometry alone, via BEV features and self-supervised anomaly detection on LiDAR, remains competitive without visual semantics [51]. TRIP instead hardens the geometric pipeline, computing a steppability risk map from a surfel representation and using steppability itself to weight terrain completion and reject temporally inconsistent measurements [78].

Humanoid work is anchored in geometry still more strongly. Mobility maps for iCub [38] replace the usual hand-tuned threshold table with Gaussian Process Regression, so that slope and roughness scores interpolate into configurations never enumerated by hand. Gholami *et al.* [33] reject learning altogether, building slopeness, roughness and empty-space indices whose explicit treatment of *voids* addresses a failure mode elevation maps handle poorly, with an adaptive ellipsoidal boundary converting the indices into locomotion-mode decisions. Yamamoto *et al.* [32] update a stochastic Elevation-and-Normal Grid each control cycle, avoiding storage and reclustering of raw point clouds. This is an early recognition that for a biped the map update rate must match the control loop, not the perception loop.

A recurring bipedal refinement replaces the dense grid by planar regions, which align with the flat finite-area contacts of a humanoid foot. HTCplanner [41] makes the trade-off explicit, using convex planar regions for footstep placement and body-obstacle checking while retaining a heightmap for swing-foot trajectories, since the volume swept between contacts is not captured by the contact surfaces. PolygMap [42] specializes this to staircases, eroding each polygon into foothold candidates safe with respect to the foot's finite footprint.

The limitation is well documented: geometry conflates appearance of support with actual support. Tall grass looks impassable yet is trivially traversable, while a smooth surface may hide ice, soft soil or a collapsing crust. The first case is addressed by regressing the true support surface beneath vegetation [67]. The second motivates Section VI-D . Bipeds are more exposed to both, since one unsupported contact is unrecoverable, yet this has been managed by conservative contact selection rather than richer semantics.

B. Foothold-, Stance-, and Stability-Aware Traversability

A distinct family, overwhelmingly bipedal in origin, redefines traversability as the availability and quality of admissible contacts, and more recently as the predicted stability of the gait they induce. Section VI-A asks whether a cell is flat enough. This family asks whether a robot with this reach, foot geometry and balance controller can place and exploit a contact there.

The earliest formulation defines traversability as the number of useful footstep combinations for a torso transition, regresses it from environment features, and uses it to bias a contact-space planner's heuristic, taming the branching factor introduced by palm contacts [25]. The follow-up converts the same

TABLE II: Terrain representation families across the quadruped and bipedal literature. Shaded cells mark families with no surveyed counterpart on that side. circled markers identify the components combined in the open direction stated in the last row.

Representation family	Quadruped literature	Humanoid / bipedal literature	What differs
Geometry-centred maps elevation, slope, occupancy	One layer among several, consumed by an explicit planner	A small robot-centric height patch used as the whole exteroceptive state of a learned policy	*The only shared substrate, but a different consumer on each side*
Contact-surface primitives planar regions, foot shape	*No counterpart.* A point-foot abstraction with dense foothold choice	Planar regions, polygonal stair maps, fitted normals, contactable-surface sets ❶	*Terrain convolved with a large polygonal sole. The two meet only at segment level*
Multi-layer cells, uncertainty distributional cells	Inclination, roughness and stiffness per cell. Distribution over *terrain properties* ❷	Elevation, normal and measurement std. Distribution over *geometry itself* ❷	*Both became distributional, but over different quantities*
Semantic and physical-property maps class, friction, compliance	Terrain class, friction, stiffness, collapsibility, vegetation pliability ❸	*No counterpart.* The property is absorbed into the controller, never written into the map	*Widest open gap, although a biped slips on low friction far more readily*
Proprioceptive, experience maps how terrain feels to traverse	Slippage, cost of transport, stability margin	Predicted instability, i.e. fall risk, indexed on the elevation map	*Same principle, different failure semantics: degraded performance vs. episode end*
Embodiment- and action-conditioned a function, not a field	Forward dynamics, traction given a commanded velocity, critic value as cost	Footstep-combination counts, foothold and stance scores, capture-region limits ❹	*Conditioned on a continuous velocity vs. a discrete contact configuration*
Volumetric and continuous fields off the ground plane	Multi-height occupancy slices, Gaussian-splat and LiDAR signed distance fields	Whole-body potential fields, ESDF at sampled footsteps, torso-collision exclusion	*Convergence from opposite causes: penetrable clutter vs. a tall narrow body*
Implicit and learned latent no explicit map	Self-supervised terrain embeddings. Novelty from reconstruction error or hypersphere	Height-map reconstruction, stability-gated BEV features, depth-to-action policies	*Latents encode terrain identity vs. recover geometry*
Open direction. A representation carrying **semantic and physical terrain properties** ❸ with **calibrated uncertainty** ❷, at **foothold resolution** ❶ and with the **balance-aware conditioning** ❹ that bipedal locomotion requires.			

regressor into a planner-selection mechanism, so that cheap planning-from-scratch handles benign path segments while an expensive retrieve-and-adapt motion library is reserved for difficult ones [26]. An under-exploited role: traversability as a signal for allocating computational effort rather than a cost to minimise.

McCrory et al. [39] encode the gait structure into the graph search by sampling the height map around nominal footholds at a predefined stance width. This yields two measures: foothold traversability, the availability of a single support, and stance traversability, which extends the same test to a paired configuration across a graph edge. This alternation is something no quadrupedal score needs to represent. A contour cost further penalizes lateral progression during elevation change, since bipeds ascend and descend more reliably along the steepest gradient [20]. Yamamoto and Sugihara [32] arrive from the control side, constraining each placement to the intersection of the traversable region, the reachable region and the one-step capture region of the current center-of-mass state. Traversability is here inseparable from capturability, and must be produced within tens of milliseconds, which rules out most learned perception pipelines.

STATE-NAV [21] is the most explicitly learning-based member, and the bipedal analogue of Section VI-E. Its TravFormer network predicts bipedal instability from a 2.5D elevation patch and a commanded velocity, supervised by body-to-stance-foot angle, a signal selected for its correlation with fall-over risk. Traversability is reported not as a scalar cost but as a stability-aware command velocity, with aleatoric uncertainty handled through Value at Risk. This resolves an ambiguity scalar cost maps leave open, since terrain is rarely traversable or not but traversable at a speed, and makes the terrain-controller coupling explicit, like value-function-derived costs [23] but with an interpretable physical quantity in place of a learned critic. It is also where the two literatures make direct contact, since quadrupedal foothold scores predict bipedal instability poorly, the clearest evidence that traversability labels do not transfer across morphologies.

Foothold reasoning is increasingly embedded inside learned controllers rather than external planners, through rewards on sole-terrain overlap [28], terrain edge detection paired with foot volume points [54], and terrain-optimal landing targets from an elevation map [36], discussed in Section VI-G.

C. Semantic and Appearance-Based Methods

A third family predicts traversability from appearance, either by classifying terrain into semantic categories with associated costs or by regressing pixel-wise scores. Early work combined texture recognition, semantic segmentation and obstacle detection from stereo imagery for footstep planning [60]. Lightweight segmentation [79] later proved sufficient for path following on constrained platforms, once paired with trajectory compensation to smooth over segmentation failures [57].

Semantic labels are only a proxy, since the class-to-cost mapping must be supplied a priori and transfers across neither robots nor conditions. Several systems therefore treat semantics as one input among several. AMCO [35] couples a vision-based general-knowledge cost map with proprioceptive maps and down-weights vision when image quality degrades. VERN [65] separates pliable from non-pliable vegetation

by few-shot siamese classification, encoding the distinction with confidence-dependent inflation. TOP-Nav [55] grounds its classifier in motion evaluations and corrects vision-based estimates online with proprioceptive experience. CAT [77] instead makes the class-to-cost mapping robot-conditioned, generating a per-platform capability vector that reshapes the semantic cost for each robot.

The near-absence of this family from humanoid work is one of the clearest asymmetries we observe. Where semantics appear in bipedal systems they are geometric rather than material: PolygMap [42] labels staircase polygons rather than materials, and others separate ground from obstacles by plane fitting rather than class [63]. To our knowledge no humanoid system distinguishes grass from gravel from ice for foothold selection, although a low-friction contact is strictly more consequential for a biped. Bipedal research has concentrated on discrete-support terrain such as stairs, beams and gaps, where geometry genuinely dominates, and the cost of a fall has discouraged deployment in the vegetated and deformable outdoor settings where semantics pay off. A notable trend is the replacement of task-specific CNN backbones by dense foundation-model features, discussed in Sections VI-E and VI-I.

D. Proprioceptive and Terrain-Property Estimation Methods

Legged robots possess a sensing channel unavailable to most wheeled platforms, the leg itself, which measures slippage, sinkage, compliance and resistance directly at the point of contact.

A first line uses proprioception as the sole signal. ProNav [34] reduces knee force differentials, hip position variations and battery current by PCA, then fits Gaussian stability ellipses per terrain-gait combination for online gait selection and crash prediction without any camera. Terrain has also been classified into physically grounded multi-label attributes such as roughness, slippage, softness and slope, from IMU and encoders alone [31]. The paradigm extends to perceptually degraded settings such as shadowed lunar regions, where proprioceptive estimates of elevation, slippage, cost of transport and stability are aggregated into a multi-layer gridmap [48].

A second line estimates explicit *physical parameters* and learns to predict them from vision, so that the robot anticipates rather than reacts. A proprioceptive decoder trained in simulation estimates per-foot friction and stiffness, distilled into a visual network for dense pixel-wise prediction [49]. See-Touch-Predict [46] makes the interaction *active*, with a learned probing policy that presses and rubs the terrain, propagating its friction estimates to visually similar terrain. Active interaction also drives the probing arm measuring collapsibility as a normalised force deficit relative to hard ground [22], a capability recent load-bearing assessment work continues to develop [80]. Ewen et al. [44] contribute the probabilistic treatment, estimating terrain classes and friction distributions jointly and in closed form from RGB-D, then fusing tactile friction with visual semantics through conjugate Bayesian updates, so that a single touch corrects a misclassification before the robot commits to a gait [47].

The common thread is the redefinition of traversability from a visual category into a measurable physical quantity, both robot-relevant and self-supervisable. The humanoid literature again diverges: bipedal systems exploit interaction forces for disturbance rejection rather than terrain identification, injecting virtual wrenches during training so the deployed policy absorbs payload perturbations without force sensing [36], or crossing low-friction and deformable terrain by denoising proprioceptive histories rather than estimating friction [50]. This gives robustness without an interpretable terrain model, which permits surviving a hazard but not planning around it. Transfer is further complicated because a biped cannot afford an exploratory probe without first establishing double support.

E. Self-Supervised and Experience-Based Learning

Manual annotation is expensive, ambiguous and robot-agnostic. The dominant modern paradigm therefore generates labels from the robot's own experience, treating terrain crossed successfully as traversable and grading the cost by the quality of that traversal.

Wellhausen et al. [68] pioneered the scheme, projecting footholds into camera images and regressing a force-derived ground reaction score pixel-wise. WVN [17] made it *online*, supervising a lightweight head on frozen foundation-model features with velocity-tracking discrepancy, which allows adaptation to a new environment within minutes of demonstration. STERLING [70] learns terrain representations from unconstrained unlabeled experience by aligning visual and inertial-proprio-tactile embeddings, while SALON [72] regresses proprioceptive roughness onto visual features in a BEV map with Gaussian-process uncertainty. Zhang et al. [23] close the loop with control, deriving the signal from the *value function* of a learned locomotion controller so that the cost map reflects what that controller can handle, and TOP-Nav [55] uses the same critic-based signal as an online proprioception advisor.

The supervisory signal is what distinguishes members of this family. Several quadruped options recur: ground reaction forces [68], velocity-tracking error [17], [72], multimodal representation alignment [70], and controller value functions [23], [55]. Bipeds instead use the body-to-stance-foot angle [21]. This signal was chosen because the ones favored in the quadrupedal literature correlated less well with fall-over risk. The paradigm therefore transfers across morphologies but its *labels* do not. The pipeline is reusable while the supervisory quantity must be re-derived per platform class.

Supervision has also been decoupled from the robot entirely. Egocentric *human* walking videos are annotated automatically by prompting a segmentation model with SLAM-estimated footsteps [43], and STEPP [45] learns the distribution of human-traversed terrain features from pose-projected embeddings. That both supervise from human bipedal gait while deploying on quadrupeds is a notable irony, and points to an opportunity for humanoids, whose contact schedule matches the demonstrator far more closely. Continual variants address the sequential nature of deployment, pairing self-supervised annotation with an incremental scene-aware memory against

catastrophic forgetting [71], or clustering VAE-encoded terrain patches so that traversability follows from similarity to accumulated experience [30]. COTRATE [81] unifies these threads into a robot-agnostic continual framework. It embeds synchronized inertial and proprioceptive series into a latent space and defines traversability as the normalized cosine similarity to a short calibration sequence collected on smooth reference terrain, capturing relative energetic and dynamic difficulty without hand-designed slip or vibration thresholds. Scores are averaged within foundation-model segments [82] to form pseudo-labels for a lightweight visual decoder. Extension to bipeds remains untested.

F. Uncertainty-Aware and Anomaly-Detection Formulations

Self-supervision from experience produces positive labels only, since the robot rarely observes catastrophic terrain, and a discriminative classifier so trained becomes overconfident out of distribution. Two complementary responses have emerged.

The first reframes traversability as *anomaly detection*. Wellhausen et al. [56] learn the distribution of safe terrain from positive multi-modal samples using autoencoders, Deep SVDD and normalizing flows, flagging anything unfamiliar as risky. The same reconstruction-loss principle is applied to foundation-model features [45]. Variants construct a positive hypersphere in latent space for LiDAR features [51], embed traversability decoding in a variational latent model [61], and use distributional divergence to detect novel scenes [71].

The second quantifies uncertainty explicitly and propagates it into planning. EVORA [40] is the most complete instance, modeling traction as a full distribution: aleatoric uncertainty by evidential deep learning over discretized traction values, epistemic uncertainty by density estimation in feature space, with a CVaR planner consuming worst-case expected traction while penalising low-density regions. Uncertainty also acts as a fusion weight, with per-sensor reliability modulating the edge weights of a graph neural network predicting trajectory success [69], image quality arbitrating between visual and proprioceptive cost maps [35], and full friction distributions supporting risk-aware planning [44].

Bipedal work has adopted the risk-sensitive formulation but not the epistemic machinery. STATE-NAV [21] applies Value at Risk to predicted instability, the counterpart of EVORA's CVaR treatment of traction [40], but models no epistemic uncertainty and detects no OOD terrain. Elsewhere uncertainty is handled through sensor modeling rather than inference. DPL [53] calibrates a stochastic depth noise model so the policy trains on realistically corrupted observations, and FocusNav [37] gates out distal information when stability degrades, acknowledging implicitly that far-field predictions become unreliable precisely when they are least affordable to act upon. Since OOD terrain is the dominant cause of catastrophic bipedal failure, transferring this tooling to humanoids appears among the most valuable open directions.

G. Predictive and Policy-Implicit Formulations

A further family builds no explicit traversability map. Traversability emerges implicitly, either as the predicted outcome of candidate actions or as a latent competence internal to a learned locomotion policy.

Learned forward dynamics: The first variant learns a *forward dynamics model* of candidate actions, letting traversability emerge as predicted failure risk. Kim et al. [74] condition such a model on 2D LiDAR scans and proprioceptive history, coupled with an informed trajectory sampler and sampling-based MPC. Roth et al. [83] extend this to perceptive, platform-aware models in which a recurrent network consuming height scans and proprioception predicts future poses and failure risk for an MPPI [84] planner [52]. Kang et al. [18] highlight the estimation side, showing that on soft terrain reliable mapping requires slip-aware leg odometry fused with IMU, LiDAR and visual odometry, with per-cell stiffness from tactile and visual cues.

Perceptive locomotion policies: The second variant, dominant in recent humanoid work, folds perception into an end-to-end reinforcement-learning controller. The question of traversability is never posed: the policy either succeeds or fails, and the level of terrain competence depends on the training distribution. PIM [27] integrates terrain perception into state estimation itself, concatenating elevation samples with proprioceptive history to predict the next proprioceptive state, while HPC [50] distills a teacher into a student that denoises observation histories to recover privileged oracle states. BeamDojo [28] targets sparse footholds with a double-critic architecture decoupling dense locomotion rewards from sparse foothold rewards, and a curriculum in which the robot explores on flat ground while receiving the foothold rewards of the true terrain, resolving the early-termination problem that otherwise makes stepping stones and beams unlearnable. Hiking in the Wild [54] pushes the opposite way, showing that a single-stage policy from raw depth to joint actions suffices given a backbone able to absorb the diversity of terrain skills.

Two systems deserve mention for how they treat the locomotion-navigation interface. Compton et al. [29] make the training reference trajectories *terrain-consistent*, projecting desired footsteps onto valid footholds inside the training loop. This removes the need for external whole-body references at inference and exposes a clean SE(2) velocity interface any planner can drive. It is the mirror image of the stability-aware command velocity [21]: one constrains the velocity a planner may request, the other guarantees the velocity a controller will maintain. Together they define a contract between estimation and locomotion that the quadrupedal literature leaves implicit. Xue et al. [24] address a problem with no quadrupedal counterpart, whole-body collision avoidance for a tall articulated body indoors. Robot-obstacle relationships become continuous differentiable gradient fields, turning sparse collision penalties into a dense anticipatory reward and enabling crouching, hurdling and squeezing that a footprint-based cost map cannot express. They argue this abstraction acts as a low-pass perceptual filter with negligible sim-to-real gap, a claim worth contrasting with DPL [53], which closes the same gap from the other end by synthesising sensor-accurate images and reconstructing local heightmaps from egocentric depth, and with TACT-ful [36], which trains on privileged elevation maps while the deployed actor consumes only noisy depth.

The strength of this subfamily is that traversability is necessarily defined relative to the controller that must execute it, the concern motivating the value-function costs of Section VI-E [23], [55]. Its weakness is that the resulting competence is opaque, non-transferable and unavailable to any planner beyond the policy's perceptual horizon.

H. Planning-Coupled and Full-Stack Navigation Systems

A final systems-oriented family treats traversability as an input to be produced fast enough, and in the right format, for a complete navigation stack. Donca et al. [63] give the most conventional bipedal instance, segmenting depth point clouds into ground and obstacles, projecting them into layered global and local 2D costmaps at different resolutions and update rates, and modeling the humanoid as a differential-drive system to suppress lateral oscillations. The design is instructive for what it discards, since reducing a biped to a 2D footprint with a velocity interface inherits three decades of wheeled navigation infrastructure at the cost of any reasoning about footholds or terrain properties. HTCplanner [41] and PolygMap [42] occupy the middle ground, keeping representations rich enough for footstep and swing-foot planning within the compute budget of a look-and-step gait.

iWalker [64] is the most architecturally distinctive of the group. Instead of separating a learned planner from a model-based controller, it uses imperative learning, a self-supervised neuro-symbolic scheme. Obstacle avoidance is expressed through a blurred signed distance field sampled at predicted footstep locations. Because supervision comes from the models rather than labels, it needs far less data than the policies of Section VI-G. FocusNav [37] addresses what a local navigator should attend to, predicting waypoints backwards from the goal and using them as queries to anchor spatial cross-attention along the path, with a stability-aware gate suppressing distal features when balance degrades. Together these illustrate that for bipeds the binding constraint is frequently not the fidelity of the estimate but the latency and interface at which it is delivered, made concrete by the millisecond-scale foot placement of [32] against the far longer planning times of [39], [20].

I. Foundation-Model and Vision-Language Approaches

The most recent family leverages large pretrained models in two roles. As *feature extractors*, frozen self-supervised vision transformers [85], [86] supply dense semantically rich embeddings on which lightweight traversability heads train with little robot data, a design underlying WVN [17], STEPP [45], cross-modal physical-parameter prediction [49], IN-Sight [61] and SALON [72]. Promptable segmentation models [87], [88], [89], [90] automate label generation in egocentric video annotation [43], IMOST [71], Splatblox [73] and TANGO [58].

As *reasoners*, vision-language models bring open-vocabulary semantics and morphology-aware judgment. SARO [75] uses a VLM as a high-level state machine decomposing 3D terrain crossing into verified sub-tasks executed by an RL locomotion policy. CATNAV [62] performs zero-shot risk inference by querying a VLM *conditioned on the robot's physical morphology*, mitigating latency with a visuosemantic cache and projecting risks into a costmap for a sampling-based planner whose final trajectory an LLM selects under natural-language constraints. CAT bakes the same morphology awareness into a trained predictor instead of a runtime query [77]. It fuses frozen DINOv3 [82] features from RGB and depth with a CLIPSeg [89] semantic map, then uses a vision-language model to turn a language description of the platform into a per-class capability vector that suppresses unusable terrain before a decoder produces a dense score. The same RGB-D frame therefore yields different maps for a legged and a wheeled robot. It is the training-time counterpart of CATNAV [62], which queries and caches that judgment zero-shot. TANGO [58] composes promptable segmentation, feature matching, contrastive language-image embeddings [91] and monocular depth into a fully zero-shot RGB-only pipeline. In parallel, neural rendering representations are being adapted to traversability, with certifiably safe corridors planned through Gaussian-Splatting maps [59] and language-derived terrain costs fused with LiDAR geometry into a traversability-aware splatting volume [73].

The promise here is generalization without robot-specific data collection. The open challenges are inference latency, the grounding of language-level judgments in robot-specific dynamics, and the absence of uncertainty quantification comparable to Section VI-F. The family is also, at the time of writing, entirely quadrupedal and wheeled, since we are aware of no humanoid traversability system reasoning in an open vocabulary. The morphology-conditioned risk queries of CATNAV [62] suit a biped conceptually well, whose feasibility constraints are far more sensitive to body geometry, and their latency budget is compatible with the meter-scale look-ahead of Section VI-H, if not with the control rates of Section VI-G.

VII. Evaluation

Traversability has no universally accepted ground truth. A terrain patch is traversable only relative to a robot, a gait, a speed and a task. This single fact shapes almost every evaluation in the surveyed literature, and it explains why the field has converged on metrics rather than on benchmarks. Authors adopt one of three strategies, often in combination: they substitute a proxy ground truth and score prediction quality, they measure the behavior of the full navigation stack, or they demonstrate the system qualitatively. This section reviews what is measured, against what, at what scale, and what the resulting evidence does and does not establish.

A. Proxy ground truth and its limits

Four proxies dominate. **Semantic labels.** GSAT [51] converts RELLIS-3D [92] and DITER++ [93] terrain classes into normal versus anomalous categories. TRIP [78] derives ground truth from SemanticKITTI [94] labels and from simulated terrain meshes in the Quadruped Robot Challenge. The authors of GSAT explicitly acknowledge the mismatch: irregular grass is

labeled traversable, and DITER++ required manually defined regions because semantic labels are unavailable.

Simulator ground truth. EVORA builds synthetic 3D maps with known traction distributions and explicit out-of-distribution masks, using 5 training, 20 test and 40 OOD environments [40]. Ewen *et al.* [44] use class-level friction distributions from simulation. Both gain controlled ground truth at the cost of simplified terrain physics. EVORA's [40] benchmark uses only dirt and vegetation and applies the same traction distribution to linear and angular motion.

Manual annotation. STEPP [45] labels 500 unseen forest images, COTRATE [81] hand-labels 2,450 test images plus grid maps, and the anomaly-detection work of Wellhausen *et al.* [56] uses roughly 500 positive and negative samples per test run.

Physical measurement. See-Touch-Predict [46] validates friction estimates against manually measured coefficients spanning $\mu = 0.17$ to $\mu = 1.03$. Chen *et al.* [49] go further with a digital twin, replaying recorded robot states in simulation and comparing joint position, base orientation and foot-slippage error; the result is an admissible friction *interval* rather than a value, 0.0–0.2 for a slippery whiteboard and 0.3–1.0 for ordinary ground.

Three forms of circularity recur, where circularity means that the test is contaminated by the method or its labels, so a good score no longer certifies independent performance. STEPP [45] tunes its 0.35 reconstruction-loss threshold on the same 500 labeled images used to report accuracy, so the figure is not an independent fixed-threshold estimate. In [47], the strongest numerical results depend on tactile friction readings sampled from ground-truth labels, which is not representative of noisy real sensing. A third, subtler mismatch appears in [30]: the benchmark ground truth is static and semantic while the method's definition of traversability is dynamic and experience-dependent, so precision–recall scores penalize correct conservative behavior on steep unvisited terrain.

B. Metrics Families

Table III groups the metrics observed across the quadruped literature. The spread is wide, but the underlying question is always one of three: is the prediction right, is the map right, or does the robot arrive.

C. Success rate is common but not comparable

Success rate is the closest thing the field has to a shared currency. It is also defined differently almost everywhere. TOP-Nav [55] requires reaching within 0.5 m of the goal in 20 s. IN-Sight [61] requires 1 m within 500 iterations, TANGO [58] 1 m within 500 steps, and the forward-dynamics planner of Kim *et al.* [74] 0.6 m without collision. The vegetation-navigation family adds semantic conditions: a run fails if the robot freezes, collides, or contacts non-pliable vegetation [65], [69], [73]. STERLING [70] makes the definition preference-based, failing any trial in which the robot traverses terrain ranked below the operator demonstration. Identical numbers therefore describe different achievements, and cross-paper comparison of raw success percentages is not meaningful.

TABLE III: Metric families used in quadruped traversability evaluation.

Family	Typical metrics
Classification	Accuracy [31], per-class accuracy [68], precision, recall, F1 [51], IoU [43], AUROC [56]
Regression / distribution	MAE [49], RMSE, KL divergence [44], squared Earth Mover's Distance [40], negative log-likelihood
Map reconstruction	Mean absolute height error, mean traversable terrain error [78], elevation RMSE [48], [67], 2D/2.5D mIoU
Task completion	Success rate, freezing rate, normalised trajectory length, SPL [61], DTW [74], collision count, false positive/negative rate [65]
Effort and comfort	Power, motor current, vibration cost [34], [35], cost of transport [48], IMU energy density, path effort, effort-weighted path length [81]
Runtime	Processing time, publishing rate [19], inference latency [43], query count [62]

Secondary metrics partially compensate. Freezing rate and normalized trajectory length separate cautious behavior from failure [65], [73]. False positive and false negative rates distinguish unsafe optimism from excessive caution. [65], [69]. SPL [61] and DTW [74] capture path efficiency rather than mere arrival. Human interventions appear where autonomy is partial: WVN [17] reports minor and major interventions over runs of 0.55, 0.5 and 1.4 km , STERLING [70] required two interventions in its three-mile deployment , and SALON [72] counts safety-driver interventions as its primary safety measure.

D. Effort as a stand-in for terrain difficulty

A growing group of works evaluates not whether the robot arrives but how much it suffers on the way. ProNav and AMCO report mean power consumption, vibration cost from hip-joint deviation, and IMU energy density from squared accelerations [34], [35]. COTRATE [81] defines path effort from average motor current and normalizes it by distance into an Effort-Weighted Path Length. TOP-Nav [55] adds average energy consumption and unstable time based on excessive roll or pitch. The planetary mapping framework aggregates cost of transport, total slip events and the GIIM/GIAM gravito-inertial margins into per-route summaries, and observes that spatial maps expose localized hazards that route averages conceal [48].

E. Baselines: a fragmented but recurring set

No standard baseline suite exists. A small set of comparators nonetheless recurs often enough to act as an informal benchmark: WVN [17] (adopted as a baseline by [71], [81], [30], [72], [49]), GA-Nav [95] (by [70], [65], [69], [34], [35], [55], [73], [72]), the geometric planner of Wermelinger et al. [96] (by [65], [69], [73]), DWA [97] (by [65], [69], [73]) and the Spot RGB-D D* planner of [98] (by [34], [35], [65]).

Beyond this cluster, evaluation is dominated by internal ablation. Several otherwise strong papers report no external quantitative baseline at all. The vision-based perception framework of Sun *et al.* [60] discusses LBP, SIFT and SURF but

TABLE IV: Reported evaluation scale of navigation-level systems.

Method	Scale
STERLING[70]	6 envs., 5 trials each; one operator's preferences; 3-mile qualitative run
VERN [65]	4 scenarios, 10 trials per method
GrASPE[69]	4 scenarios, $\geq$10 trials per method
AMCO[35]	4 scenarios, 10 trials
CATNAV[62]	5 tasks, 10 trials, single session
TOP-Nav[55]	64 sim. cells $\times$ 25 trials; 5 real repetitions per method
Splatblox[73]	10 runs short-range, 3 runs per 100 m mission
SALON[72]	5 laps on a waypoint course; quadruped results teleoperated
EVORA[40]	5 indoor two-lap races; 6 outdoor trials per method
GSAT[51]	10 sim. nav. trials, one robot, one planner
SARO[75]	20 real trials per terrain; 4,096 sim. runs for the low-level policy
IN-Sight[61]	100 epochs per configuration, sim. only
TANGO[58]	36 envs., 108 episodes, sim.

reports no comparative accuracy or runtime. The Pegasus-Mini [57] pipeline compares only dataset choices and the addition of trajectory compensation. See-Touch-Predict [46] compares its feature-extraction network against retrained and augmentation-free variants rather than against prior terrain-perception systems. The anomaly-detection study compares autoencoders, Deep SVDD and Real-NVP against one another [56]. IN-Sight [61] compares against deterministic, no-RNN and VAE-only variants plus one minimal baseline. The planetary mapping framework is positioned conceptually against ProNav [34] and related proprioceptive methods without a quantitative benchmark [48]. Haddeler *et al.* [22] compare only against Tennakoon *et al.*, and describe their work as a proof of concept.

F. Statistical strength

Physical evaluation is small. Table IV lists reported scales for the navigation-level systems. Five to ten trials per scenario, one robot, and three to six hand-selected sites is the norm. Uncertainty estimates and significance testing are almost never reported.

Several papers are explicit that the missing evidence is safety evidence. FDM [52] notes that real-world collision data are absent because collecting collisions risks damaging hardware, leaving a sim-to-real gap precisely where it matters most. Splat-Nav [59] evaluates real-world safety against its own Gaussian-Splat collision geometry rather than an independent ground-truth mesh, because suitable geometry was unavailable.

G. Bipedal and humanoid evaluation

Evaluation on bipedal platforms follows a visibly different pattern (Table V). Explicit traversability estimation is evaluated by only a minority of works: planning-centric metrics dominate, chiefly planning time, planner success rate, and footstep tracking error. Simulation is the primary venue, with hardware confined to a small set of platforms.

TABLE V: Evaluation venues and primary metrics on bipedal and humanoid platforms.

Method	Primary evidence
STATE-NAV[21]	3 sim envs., 10 trials/method; 3 outdoor sites
Lin (2017)[25]	95 envs.; planning success and time
Lin (2018)[26]	2 env. types, 50 trials each
McCrory[39], [20]	Ramps, stairs, cinder blocks, stepping stones; plan time
Yamamoto[32]	2 scenarios; 10–20 ms planning; perturbation robustness
Genesio[38]	Indoor corridors; 486 ms per cloud
Transcend[33]	Slope and roughness statistics, semi-structured indoor
Donca[63]	Sim. 6×6 m, hardware 10×8 m; 0.2 m / 0.2 rad tolerance
HumanoidPF[24]	8 sim scene types; 4 real scenarios, 4/5–5/5 success
FocusNav[37]	100 episodes $\times$ 3 seeds; 15 real trials $\times$ 6 scenarios
BeamDojo[28]	5 terrain types; sim and zero-shot transfer
PIM[27]	Stairs, platforms, gaps, slopes; cross-platform transfer
DPL[53]	Stumble counts; reconstruction error
TACT-ful[36]	4,096 envs.; 10 m traversal without fall; payload tests
PolygMap[42]	3 phases; execution time, step count, tracking error
iWalker[64]	1,185 depth frames, 8 scenes; 4 unseen scenes
Hiking[54]	10 trials per terrain-gait pair; 4-minute run

The planning-oriented line reports gains in planner efficiency rather than perception quality. Learned traversability estimates improve planning success and reduce planning time over heuristic baselines [25], [26], while the Atlas pipelines report end-to-end plan times of a few seconds split between A* search and optimisation [39], [20]. Faster planners report millisecond-scale planning in simulation [32], and older mobility-map work is validated by reprojection onto RGB frames [38].

STATE-NAV [21] is the clearest case of traversability-specific evaluation on a biped, training on simulated gait cycles and comparing against manual traversability rules, quadrupedal foothold scores, and learned instability baselines. At the opposite end, Gholami and Ramirez-Serrano [33] report terrain statistics rather than navigation outcomes, characterizing slope and roughness across a semi-structured indoor space.

The adjacent perceptive-locomotion literature evaluates almost exclusively by terrain-specific success rate and fall avoidance [37], [28], [27]. Others [53], [29], [64] report improvement counts rather than rates, such as reduced stair stumbles, tracking error, or feasibility violations . Elsewhere, endurance demonstrations substitute for statistics, from hours of real-world walking to tens of meters of autonomous outdoor navigation over stairs.

H. Discussion

Five observations follow from the material above. **Two evaluation cultures coexist and rarely meet.** Perception-oriented work scores predictions against proxy labels; navigation-oriented work scores arrivals on bespoke courses. Few papers close the loop between them. Those that do are instructive:

COTRATE [81] links segmentation mIoU to path effort , GSAT [51] links anomaly F1 to collision counts in simulated navigation , and FDM [52] links collision-prediction F1 to goal-success rate. Elsewhere the connection is assumed. Several authors say so explicitly: STEPP, Kim *et al.* and the anomaly-detection study all note that their metrics measure prediction quality rather than navigation safety, completion or intervention frequency [45], [43], [56].

The absence of ground truth is a methodological constraint, not a detail. Every proxy distorts in a known direction. Semantic labels penalize robot-specific and experience-dependent definitions [30]. Simulator ground truth buys control at the cost of terrain realism [40]. Manual annotation is small and site-specific. Physical measurement is accurate but sparse, and even then produces intervals rather than values [49].

Reported gains are typically over weak or internal baselines. The field has no shared test set, no shared course, and no shared success definition. The informal cluster around WVN [17], GA-Nav [95], DWA [97] and the Wermelinger geometric planner [96] is the closest approximation, and it is used inconsistently.

Statistical power is low relative to the claims. Five to ten trials on one platform at one site cannot separate a genuine improvement from run-to-run variance, and this limitation is acknowledged in nearly every paper that performs physical experiments.

Safety, the motivating concern, is the least directly evaluated property. Collisions and falls are avoided in data collection because they damage hardware [52], so safety is inferred from success rates, freezing rates and false-positive rates rather than measured. A standardized legged traversability benchmark, with a shared terrain suite, explicit robot-conditioned ground truth, and reporting of both prediction and task metrics with variance, remains the clearest open need.

VIII. Open Challenges

Despite the progress reviewed above, several challenges still prevent traversability estimation from becoming a reliable and transferable capability for autonomous legged robots. Many locomotion-relevant terrain properties are only partially observable before interaction, while geometry and appearance alone remain insufficient to infer friction, stiffness, deformability, load-bearing capacity, sinkage, or slip risk. Although recent methods increasingly combine exteroceptive look-ahead with proprioceptive, contact, and tactile feedback, robust fusion remains challenging under occlusion, sensor degradation, calibration errors, and varying modality reliability, and interaction-derived information must support prediction rather than merely retrospective assessment.

Uncertainty and risk remain inconsistently addressed. Aleatoric and epistemic uncertainty are rarely modeled jointly, unfamiliar terrain may be assigned overconfident costs, and risk is seldom propagated coherently from cells or footholds to complete trajectories. Such propagation also requires preserving the dependence of traversability on morphology, dynamic state, gait, intended contacts, and controller capability.

Generalization remains limited despite increasing research on continual learning, online adaptation, and sim-to-real transfer. Estimators are often tied to a specific platform, sensor suite, controller, environment, or supervisory signal, and progress is further constrained by the lack of representative datasets and standardized evaluation protocols covering physical terrain properties, uncertainty, runtime, and downstream navigation performance, both within and ultimately across morphologies.

Only a few works use RL or inverse RL to construct explicit traversability models or controller-derived cost representations [99], [23]. Most RL research instead learns terrain-aware locomotion, leaving traversability implicit in the policy, reward, or value function. Open questions include how to derive reusable and interpretable costs or risks, quantify uncertainty, prevent unsafe exploration, and expose learned competence to external planners.

Finally, recent foundation-model approaches have begun to provide open-vocabulary semantic reasoning and reduce dependence on task-specific annotations, but their traversability judgments remain weakly grounded in morphology, contact feasibility, terrain physics, and controller limitations, while inference latency and calibrated uncertainty remain unresolved.

IX. Conclusions

This survey reviewed traversability analysis for legged robots. We departed from the wheeled formulation and defined traversability as the robot-dependent degree to which terrain supports safe, stable and efficient legged motion. The field was organized around sensing, terrain representation, problem formulation, estimation and evaluation, covering quadrupeds and humanoids together.

The central finding is that morphology sets the question. A quadruped keeps a support polygon after a misstep, so terrain is scored patch by patch as a field of local attributes. A humanoid walks in single support on one sole, so a patch is admissible only when a whole foot fits and the safe set is bounded by the capture region. The quadrupedal literature therefore asks what the terrain is made of. The humanoid literature asks whether a stable contact schedule exists on it. The two communities have cross-fertilized little. Physical and semantic terrain properties, calibrated uncertainty and out-of-distribution detection are routine on quadrupeds and largely absent on humanoids, even though unfamiliar terrain is the main cause of catastrophic bipedal failure. Conversely, the quadrupedal literature has no counterpart to traversability defined over the contact configuration space, to stance-level scoring, or to whole-body clearance fields for crouching, hurdling and squeezing.

A second theme is the interface between estimation and control. Explicit maps are interpretable and reusable but only approximate the controller they serve. Policy-implicit traversability is exact for its controller but opaque and untransferable. Recent bipedal work occupies a deliberate middle ground and sketches a contract between perception and locomotion that the quadrupedal literature leaves implicit.

Evaluation remains the weakest link. Traversability has no accepted ground truth. Success rate is defined differently

in almost every paper. Statistical power is low. Safety, the property that motivates the problem, is the least directly measured. Progress depends on two convergences. The first is representational. Semantic and physical properties should be carried with calibrated uncertainty, resolved at foothold scale and conditioned on the robot's dynamic state. The second is methodological. A standardized legged benchmark should share a terrain suite, provide robot-conditioned ground truth and report prediction and task metrics with variance. Until then, traversability estimation for legged robots will remain a collection of well-engineered systems rather than a transferable capability.

REFERENCES

[1] A. E. Carvalho, D. Portugal, and P. Peixoto, "On terrain traversability analysis in unstructured environments: Recent advances in forest applications," *Intelligent Service Robotics*, 2025.

[2] M. Benrabah, C. O. Mousse, E. Randriamiarintsoa, R. Chapuis, and R. Aufrère, "A review on traversability risk assessments for autonomous ground vehicles: Methods and metrics," *Sensors*, 2024.

[3] C. Sevastopoulos and S. Konstantopoulos, "A survey of traversability estimation for mobile robots," *IEEE Access*, 2022.

[4] P. V. K. Borges, T. Peynot, S. Liang, B. Arain, M. Wildie, M. G. Minareci, S. Lichman, G. Samvedi, I. Sa, N. Hudson, M. Milford, P. Moghadam, and P. Corke, "A survey on terrain traversability analysis for autonomous ground vehicles: Methods, sensors, and challenges," *Field Robotics*, 2022.

[5] D. C. Guastella and G. Muscato, "Learning-based methods of perception and navigation for ground vehicles in unstructured environments: A review," *Sensors*, 2021.

[6] S. Beycimen, D. Ignatyev, and A. C. Zolotas, "A comprehensive survey of unmanned ground vehicle terrain traversability for unstructured environments and sensor technology insights," *Engineering Science and Technology, an International Journal*, 2023.

[7] F. Atas, G. Cielniak, and L. Grimstad, "From simulation to field: Learning terrain traversability for real-world deployment," *arXiv preprint arXiv:2501.06904*, 2025.

[8] C. Qin, G. P. Ruwanpathirana, S. Thilakarathna, H. Wen, Y. Ji, J. Yue, and S. K. Baduge, "A comprehensive review of quadruped robots: Vision perception, motion control, applications and challenges," *Journal of Automation and Intelligence*, 2026.

[9] Z. Wu, K. Zheng, Z. Ding, and H. Gao, "A survey on legged robots: Advances, technologies and applications," *Engineering Applications of Artificial Intelligence*, 2024.

[10] Y. Tong, H. Liu, and Z. Zhang, "Advancements in humanoid robots: A comprehensive review and future prospects," *IEEE/CAA Journal of Automatica Sinica*, 2024.

[11] W. Li, G. Yang, J. Wu, C. Pan, L. Sheng, and Q. Zhang, "A comprehensive review on humanoid robots: Perspectives from academia and industry," *ENGINEERING Information Technology & Electronic Engineering*, 2026.

[12] Z. Gu, J. Li, W. Shen, W. Yu, Z. Xie, S. McCrory, X. Cheng, A. Shamsah, R. Griffin, C. K. Liu, A. Kheddar, X. B. Peng, Y. Zhu, G. Shi, Q. Nguyen, G. Cheng, H. Gao, and Y. Zhao, "Humanoid locomotion and manipulation: Current progress and challenges in control, planning, and learning," *IEEE/ASME Transactions on Mechatronics*, 2026.

[13] T. Bin, H. Yan, N. Wang, M. N. Nikolić, J. Yao, and T. Zhang, "A survey on the visual perception of humanoid robot," *Biomimetic Intelligence and Robotics*, 2025.

[14] A. Roychoudhury, S. Khorshidi, S. Agrawal, and M. Bennewitz, "Perception for humanoid robots," *Current Robotics Reports*, 2023.

[15] C. R. de Lima, S. G. Khan, M. Tufail, S. H. Shah, and M. R. O. A. Maximo, "Humanoid robot motion planning approaches: a survey," *Journal of Intelligent & Robotic Systems*, 2024.

[16] I.-C. Chang, X. Huang, T.-Y. Lin, S. Teng, W. Li, M. Ghaffari, J. Yi, and Y. Gu, "A survey of legged robotics in non-inertial environments: Past, present, and future," *arXiv preprint arXiv:2604.20990*, 2026.

[17] J. Frey, M. Mattamala, N. Chebrolu, C. Cadena, M. F. Fallon, and M. Hutter, "Fast traversability estimation for wild visual navigation," in *Proceedings of Robotics: Science and Systems*, 2023.

[18] P. Kang, X. Mu, W. Xu, and Y. Xu, "State estimation and traversability map construction method of a quadruped robot on soft uneven terrain," *Journal of Field Robotics*, 2023.

[19] H. Xie, C. Cui, X. Zhong, X. Zhong, and Q. Liu, "Real-time support terrain mapping and terrain adaptive local planning for quadruped robots," *IEEE Robotics and Automation Letters*, 2024.

[20] S. McCrory, B. Mishra, R. J. Griffin, J. E. Pratt, and H. E. Sevil, "Bipedal navigation planning over rough terrain using traversability models," in *SoutheastCon 2023*, 2023.

[21] Z. Yoon, L. Y. Zhu, J. Lu, L. Gan, and Y. Zhao, "State-nav: Stability-aware traversability estimation for bipedal navigation on rough terrain," *IEEE Robotics and Automation Letters*, 2025.

[22] G. Haddeler, M. Y. M. Chuah, Y. You, J. Chan, A. H. Adiwahono, W.-Y. Yau, and C.-M. Chew, "Traversability analysis with vision and terrain probing for safe legged robot navigation," *Frontiers in Robotics and AI*, 2022.

[23] H. Zhang, Z. Li, X. Zeng, L. Smith, K. Stachowicz, D. Shah, L. Yue, Z. Song, W. Xia, S. Levine, K. Sreenath, and Y.-H. Liu, "Traversability-aware legged navigation by learning from real-world visual data," *IEEE Transactions on Robotics*, 2026.

[24] H. Xue, S. Liang, Z. Zhang, Z. Zeng, Y. Liu, Y. Lian, J. Wang, Q. Liu, X. Shi, and L. Yi, "Collision-free humanoid traversal in cluttered indoor scenes," *IEEE Robotics and Automation Letters*, 2026.

[25] Y.-C. Lin and D. Berenson, "Humanoid navigation in uneven terrain using learned estimates of traversability," in *2017 IEEE-RAS 17th International Conference on Humanoid Robotics (Humanoids)*, 2017.

[26] ——, "Humanoid navigation planning in large unstructured environments using traversability-based segmentation," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.

[27] J. Long, J. Ren, M. Shi, Z. Wang, T. Huang, P. Luo, and J. Pang, "Learning humanoid locomotion with perceptive internal model," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[28] H. Wang, Z. Wang, J. Ren, Q. Ben, T. Huang, W. Zhang, and J. Pang, "Beamdojo: Learning agile humanoid locomotion on sparse footholds," in *Proceedings of Robotics: Science and Systems*, 2025.

[29] W. D. Compton, Z. Olkin, and A. D. Ames, "Terrain consistent reference-guided rl for humanoid navigation autonomy," *arXiv preprint arXiv:2605.15517*, 2026.

[30] M. Á. de Miguel, J. Beltrán, J. S. Cely, F. Martín, J. C. Manzanares, and A. García, "I move therefore i learn: Experience-based traversability in outdoor robotics," *IEEE Access*, 2026.

[31] Y. Zhang, B. Huang, M. Hong, C. Huang, G. Wang, and M. Guo, "A terrain classification method for quadruped robots with proprioception," *Electronics*, 2025.

[32] T. Yamamoto and T. Sugihara, "Responsive navigation of a biped robot that takes into account terrain, foot-reachability and capturability," *Advanced Robotics*, 2021.

[33] A. Gholami and A. Ramirez-Serrano, "Terrain traversability via sensed data for robots operating inside heterogeneous, highly unstructured spaces," *Sensors*, 2025.

[34] M. B. Elnoor, A. J. Sathyamoorthy, K. Weerakoon, and D. Manocha, "Pronav: Proprioceptive traversability estimation for legged robot navigation in outdoor environments," *IEEE Robotics and Automation Letters*, 2024.

[35] M. B. Elnoor, K. Weerakoon, A. J. Sathyamoorthy, T. Guan, V. Rajagopal, and D. Manocha, "Amco: Adaptive multimodal coupling of vision and proprioception for quadruped robot navigation in outdoor environments," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[36] T. Ly, T.-D. Dang, C. Le, T.-D. Do, P. T. Dat, C. T. Trinh, V. A. Ngo, and A. T. Le, "Tact-ful: Multi-channel terrain affordance and compliance training for payload-robust perceptive humanoid locomotion," *arXiv preprint arXiv:2606.20645*, 2026.

[37] Y. Zhang, J. Ma, L. Yan, Z. Cao, Y. Zhang, H. Li, and Y. Gao, "Focusnav: Spatial selective attention with waypoint guidance for humanoid local navigation," *arXiv preprint arXiv:2601.12790*, 2026.

[38] N. Genesio, T. Abuhashim, F. Solari, M. Chessa, and L. Natale, "Mobility map computations for autonomous navigation using an rgbd sensor," *arXiv preprint arXiv:1610.01326*, 2016.

[39] S. McCrory, B. Mishra, J. An, R. J. Griffin, J. E. Pratt, and H. E. Sevil, "Humanoid path planning over rough terrain using traversability assessment," *arXiv preprint arXiv:2203.00602*, 2022.

[40] X. Cai, S. Ancha, L. Sharma, P. R. Osteen, B. Bucher, S. Phillips, J. Wang, M. Everett, N. Roy, and J. P. How, "Evora: Deep evidential traversability learning for risk-aware off-road autonomy," *IEEE Transactions on Robotics*, 2024.

[41] F. Chen, R. Wan, N. Zheng, R. Jiao, J. Jiang, and B. Zhou, "Htcplanner: A real-time hierarchical traversability and crossability-based planner for bipedal robots," in *AI Enabled Robotic Loco-Manipulation*, Q. Li, M. Xie, M. O. Tokhi, and M. F. Silva, Eds., 2026.

[42] B. Li, N. Wang, T. Zhang, Z. He, and Y. Wu, "Polygmap: A perceptive locomotion framework for humanoid robot stair climbing," *arXiv preprint arXiv:2510.12346*, 2025.

[43] Y. Kim, J. H. Lee, C. Lee, J. Mun, D. Youm, J. Park, and J. Hwangbo, "Learning semantic traversability with egocentric video and automated annotation strategy," *IEEE Robotics and Automation Letters*, 2024.

[44] P. Ewen, A. Li, Y. Chen, S. Hong, and R. Vasudevan, "These maps are made for walking: Real-time terrain property estimation for mobile robots," *IEEE Robotics and Automation Letters*, 2022.

[45] S. Ægidius, D. Hadjivelichkov, J. Jiao, J. Embley-Riches, and D. Kanoulas, "Watch your stepp: Semantic traversability estimation using pose projected features," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[46] H. Lin, H. Li, and Y. Gao, "See-touch-predict: Active exploration and online perception of terrain physics with legged robots," *IEEE Robotics and Automation Letters*, 2025.

[47] P. Ewen, H. Chen, Y. Chen, A. Li, A. Bagali, G. Gunjal, and R. Vasudevan, "You've got to feel it to believe it: Multi-modal bayesian inference for semantic and property prediction," *Robotics: Science and Systems*, 2024.

[48] A. Sánchez-Delgado, J. C. V. Soares, V. Barasuol, and C. Semini, "Towards proprioceptive terrain mapping with quadruped robots for exploration in planetary permanently shadowed regions," in *2025 International Conference on Space Robotics (iSpaRo)*, 2025.

[49] J. Chen, J. Frey, R. Zhou, T. Miki, G. Martius, and M. Hutter, "Identifying terrain physical parameters from vision - towards physical-parameter-aware locomotion and navigation," *IEEE Robotics and Automation Letters*, 2024.

[50] W. Sun, B. Cao, L. Chen, Y. Su, Y. Liu, Z. Xie, and H. Liu, "Learning perceptive humanoid locomotion over challenging terrain," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[51] D. Cho, M. Park, J. Lee, G. Yang, and Y. Cho, "Gsat: Geometric traversability estimation using self-supervised learning with anomaly detection for diverse terrains," 2026.

[52] P. Roth, J. Frey, C. Cadena, and M. Hutter, "Learned perceptive forward dynamics model for safe and platform-aware robotic navigation," in *Proceedings of Robotics: Science and Systems*, 2025.

[53] J. Sun, G. Han, P. Sun, W. Zhao, J. Cao, J. Wang, Q. Zhang, and Y. Guo, "Dpl: Depth-only perceptive humanoid locomotion via realistic depth synthesis and cross-attention terrain reconstruction," *IEEE Robotics and Automation Letters*, 2026.

[54] S. Zhu, Z. Zhuang, M. Zhao, K.-Y. Lee, and H. Zhao, "Hiking in the wild: A scalable perceptive parkour framework for humanoids," *arXiv preprint arXiv:2601.07718*, 2026.

[55] J. Ren, Y. Liu, Y. Dai, J. Long, and G. Wang, "Top-nav: Legged navigation integrating terrain, obstacle and proprioception estimation," in *Proceedings of The 8th Conference on Robot Learning*, 2024.

[56] L. Wellhausen, R. Ranftl, and M. Hutter, "Safe robot navigation via multi-modal anomaly detection," *IEEE Robotics and Automation Letters*, 2020.

[57] G. Deng, J. Luo, C. Sun, D. Pan, L. Peng, N. Ding, and A. Zhang, "Vision-based navigation for a small-scale quadruped robot pegasus-mini," in *2021 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2021.

[58] S. Podgorski, S. Garg, M. Hosseinzadeh, L. Mares, F. Dayoub, and I. Reid, "Tango: Traversability-aware navigation with local metric control for topological goals," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[59] T. Chen, O. Shorinwa, J. Bruno, A. Swann, J. Yu, W. Zeng, K. Nagami, P. M. Dames, and M. Schwager, "Splat-nav: Safe real-time robot navigation in gaussian splatting maps," *IEEE Transactions on Robotics*, 2025.

[60] J. Sun, Y. Meng, J. Tan, C. Sun, J. Zhang, N. Ding, H. Qian, and A. Zhang, "A vision-based perception framework for outdoor navigation tasks applicable to legged robots," in *2017 Chinese Automation Congress (CAC)*, 2017.

[61] P. Schoch, F. Yang, Y. Ma, S. Leutenegger, M. Hutter, and Q. Leboutet, "In-sight: Interactive navigation through sight," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[62] A. Potnis, F. Affonso, S. Gummadi, N. K. Uppalapati, and G. Chowdhary, "Catnav: Cached vision-language traversability for efficient zero-shot robot navigation," *arXiv preprint arXiv:2603.22800*, 2026.

[63] O. A. Donca, C. Beokhaimook, and A. Hereid, "Real-time navigation for bipedal robots in dynamic environments," *arXiv preprint arXiv:2210.03280*, 2022.

[64] X. Lin, Y. Huang, T. Fu, X. Xiong, and C. Wang, "iwalker: Imperative visual planning for walking humanoid robot," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[65] A. J. Sathyamoorthy, K. Weerakoon, T. Guan, M. Russell, D. Conover, J. Pusey, and D. Manocha, "Vern: Vegetation-aware robot navigation in dense unstructured outdoor environments," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[66] T. Miki, L. Wellhausen, R. Grandia, F. Jenelten, T. Homberger, and M. Hutter, "Elevation mapping for locomotion and navigation using gpu," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[67] K. Weerakoon, A. J. Sathyamoorthy, J. Liang, T. Guan, U. Patel, and D. Manocha, *IEEE Robotics and Automation Letters*, 2023.

[68] L. Wellhausen, A. Dosovitskiy, R. Ranftl, K. Walas, C. Cadena, and M. Hutter, "Where should i walk? predicting terrain properties from images via self-supervised learning," *IEEE Robotics and Automation Letters*, 2019.

[69] K. Weerakoon, A. J. Sathyamoorthy, J. Liang, T. Guan, U. Patel, and D. Manocha, "Graspe: Graph based multimodal fusion for robot navigation in outdoor environments," *IEEE Robotics and Automation Letters*, 2023.

[70] H. Karnan, E. Yang, D. Farkash, G. Warnell, J. Biswas, and P. Stone, "Sterling: Self-supervised terrain representation learning from unconstrained robot experience," in *Proceedings of The 7th Conference on Robot Learning*, 2023.

[71] K. Ma, Z. Sun, C. Xiong, Q. Zhu, K. Wang, and L. Pei, "Imost: Incremental memory mechanism with online self-supervision for continual traversability learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[72] M. Sivaprakasam, S. Triest, C. Ho, S. Aich, J. Lew, I. Adu, W. Wang, and S. Scherer, "Salon: Self-supervised adaptive learning for off-road navigation," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[73] S. Chopra, J. Liang, G. Seneviratne, Y. Lee, J. Choi, J. An, S. Cheng, and D. Manocha, "Splatblox: Traversability-aware gaussian splatting for outdoor robot navigation," *arXiv preprint arXiv:2511.18525*, 2025.

[74] Y. Kim, C. Kim, and J. Hwangbo, "Learning forward dynamics model and informed trajectory sampler for safe quadruped navigation," in *Proceedings of Robotics: Science and Systems*, 2022.

[75] S. Zhu, D. Li, L. Mou, Y. Liu, N. Xu, and H. Zhao, "Saro: Space-aware robot system for terrain crossing via vision-language model," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[76] Y. Chen, J. Ma, Z. Luo, Y. Han, Y. Dong, B. Xu, and P. Lu, "Learning autonomous and safe quadruped traversal of complex terrains using multi-layer elevation maps," *IEEE Robotics and Automation Letters*, 2025.

[77] G. Capezzuto, F. Tommaselli, M. P. Angarola, R. V. Godoy, and M. Becker, "Towards capability-aware traversability navigation for unstructured environments," 2026.

[78] M. Oh, B. Yu, I. M. A. Nahrendra, S. Jang, H. Lee, D. Lee, S. Lee, Y. Kim, M. K. Christiansen, H. Lim, and H. Myung, "Trip: Terrain traversability mapping with risk-aware prediction for enhanced online quadrupedal robot navigation," *arXiv preprint arXiv:2411.17134*, 2024.

[79] E. Romera, J. M. Álvarez, L. M. Bergasa, and R. Arroyo, "Erfnet: Efficient residual factorized convnet for real-time semantic segmentation," *IEEE Transactions on Intelligent Transportation Systems*, 2018.

[80] V. S. Medeiros, G. B. Dessy, T. Boaventura, M. Becker, C. Semini, and V. Barasuol, "Load-bearing assessment for safe locomotion of quadruped robots on collapsing terrain," *IEEE Robotics and Automation Letters*, 2025.

[81] J. Hindel, S. Bultmann, H. Masnavi, D. Cattaneo, and A. Valada, "Self-supervised online robot-agnostic traversability estimation for open-world environments," *arXiv preprint arXiv:2605.28442*, 2026.

[82] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa, F. Massa, D. Haziza, L. Wehrstedt, J. Wang, T. Darcet, T. Moutakanni, L. Sentana, C. Roberts, A. Vedaldi, J. Tolan, J. Brandt, C. Couprie, J. Mairal, H. Jégou, P. Labatut, and P. Bojanowski, "Dinov3," *arXiv preprint arXiv:2508.10104*, 2025.

[83] R. Dey and F. M. Salem, "Gate-variants of gated recurrent unit (gru) neural networks," in *2017 IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2017.

[84] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control using covariance variable importance sampling," *arXiv preprint arXiv:1509.01149*, 2015.

[85] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "Dino: Detr with improved denoising anchor boxes for end-to-end object detection," in *International Conference on Learning Representations*, 2023.

[86] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jégou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, "Dinov2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.

[87] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollar, and R. Girshick, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[88] X. Zhao, W. Ding, Y. An, Y. Du, T. Yu, M. Li, M. Tang, and J. Wang, "Fast segment anything," *arXiv preprint arXiv:2306.12156*, 2023.

[89] T. Lüddecke and A. S. Ecker, "Image segmentation using text and image prompts," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[90] B. Cheng, A. Choudhuri, I. Misra, A. Kirillov, R. Girdhar, and A. G. Schwing, "Mask2former for video instance segmentation," *arXiv preprint arXiv:2112.10764*, 2021.

[91] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning*, 2021.

[92] P. Jiang, P. R. Osteen, M. Wigness, and S. Saripalli, "Rellis-3d dataset: Data, benchmarks and analysis," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[93] J. Kim, H. Kim, S. Jeong, Y. Shin, and Y. Cho, "Diter++: Diverse terrain and multi-modal dataset for multi-robot slam in multi-session environments," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[94] J. Behley, M. Garbade, A. Milioto, J. Quenzel, S. Behnke, C. Stachniss, and J. Gall, "Semantickitti: A dataset for semantic scene understanding of lidar sequences," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[95] T. Guan, D. Kothandaraman, R. Chandra, A. J. Sathyamoorthy, K. Weerakoon, and D. Manocha, "Ga-nav: Efficient terrain segmentation for robot navigation in unstructured outdoor environments," *IEEE Robotics and Automation Letters*, 2022.

[96] M. Wermelinger, P. Fankhauser, R. Diethelm, P. Krüsi, R. Siegwart, and M. Hutter, "Navigation planning for legged robots in challenging terrain," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016.

[97] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, 1997.

[98] A. Koval, S. Karlsson, and G. Nikolakopoulos, "Experimental evaluation of autonomous map-based spot navigation in confined environments," *Biomimetic Intelligence and Robotics*, 2022.

[99] L. Gan, J. W. Grizzle, R. M. Eustice, and M. Ghaffari, "Energy-based legged robots terrain traversability modeling via deep inverse reinforcement learning," *IEEE Robotics and Automation Letters*, 2022.

TABLE VI: Surveyed traversability methods, ordered by year. Sensing: Ext exteroceptive, Pro proprioceptive, Multi multimodal. Representation: Elev elevation or occupancy, Cont contact surfaces, Multi multi-layer, Sem semantic or physical, Exp experience, Act action conditioned, Vol volumetric, Lat latent. Family: Geo geometry, Foot foothold and stance, Sem semantic, Prop terrain property, Self self supervised, Unc uncertainty, Pred predictive, Plan planning coupled, Found foundation model. Morphology: Q quadruped, B biped. A blank cell means the manuscript does not place that method on that axis.

Method	Year	Venue	Morph.	Sensing	Representation	Family	Code
Genesio et al. [38]	2016	arXiv	B	Ext	Cont	Geo	Y
Lin et al. [25]	2017	Humanoids	B	Ext	Cont, Act	Foot	N
Sun et al. [60]	2017	CAC	Q	Multi	Sem	Sem, Prop, Geo	N
Lin et al. [26]	2018	IROS	B	Ext	Cont, Act, Exp	Foot, Plan	N
Wellhausen et al. [68]	2019	RA-L	Q	Multi	Sem	Self, Prop, Sem, Unc	N
Wellhausen et al. [56]	2020	RA-L	Q	Ext	Lat	Unc, Self	Y
Deng et al. [57]	2021	ROBIO	Q	Ext	Sem	Sem	N
Yamamoto et al. [32]	2021	Adv. Robotics	B	Ext	Multi, Act	Geo, Foot, Plan	N
Donca et al. [63]	2022	arXiv	B	Ext	Vol	Sem, Plan	N
Ewen et al. [44]	2022	RA-L	Q, B	Ext, Multi	Multi, Sem	Prop, Unc	Y
GrASPE [69]	2022	arXiv	Q	Multi	Lat, Act	Unc, Pred, Plan	N
Haddeler et al. [22]	2022	Front. Robot. AI	Q	Multi	Elev, Sem	Geo, Prop	N
Kim et al. [74]	2022	RSS	Q	Multi	Act	Pred	Y
McCrory et al. [39]	2022	arXiv	B	Ext	Elev, Cont, Act	Foot, Plan	Y
Miki et al. [66]	2022	IROS	Q	Ext	Elev	Geo, Unc	Y
EVORA [40]	2023	T-RO	Q	Multi	Elev, Sem, Lat	Self, Unc, Prop, Plan	Y
Frey et al. [17]	2023	arXiv	Q	Multi	Cont, Sem, Lat	Self, Found	Y
Kang et al. [18]	2023	Field Robotics	Q	Pro, Multi	Multi	Pred	N
Li et al. [67]	2023	RA-L	Q	Ext, Multi	Sem	Self, Sem, Geo	Y
McCrory et al. [20]	2023	SoutheastCon	B	Ext	Elev, Cont, Act	Geo, Foot, Plan	Y
ProNav [34]	2023	RA-L	Q	Ext, Pro	Exp	Prop	N
STERLING [70]	2023	ICRA	Q	Multi	Lat	Self, Plan	N
VERN [65]	2023	IROS	Q	Ext	Sem, Vol	Sem	N
AMCO [35]	2024	IROS	Q	Multi	Exp	Sem, Unc	N
Chen et al. [49]	2024	RA-L	Q	Multi	Sem	Prop, Found	Y
Ewen et al. [47]	2024	arXiv	B	Multi	Sem	Prop	Y
IMOST [71]	2024	ICRA	Q	Multi	Lat	Self, Unc, Found	Y
IN-Sight [61]	2024	IROS	Q	Ext	Sem, Lat	Unc, Found	N
Kim et al. [43]	2024	RA-L	Q	Ext	Cont	Self, Found	N
Lin et al. [64]	2024	IROS	B	Ext	Vol	Plan	N
Long et al. [27]	2024	ICRA	B	Multi	Elev	Pred, Self, Plan	Y
SALON [72]	2024	ICRA	Q	Multi	Sem, Lat	Self, Found	Y
SARO [75]	2024	ICRA	Q	Multi	Sem, Lat	Found	N
Splat-Nav [59]	2024	T-RO	-	Ext	Vol	Found	Y
TOP-Nav [55]	2024	CoRL	Q	Multi	Exp	Sem, Self, Pred	Y
TRIP [78]	2024	arXiv	Q	Ext	Elev, Sem	Geo, Unc, Plan	N
Xie et al. [19]	2024	RA-L	Q	Ext, Multi	Elev	Geo	N
Zhang et al. [23]	2024	T-RO	Q	Ext, Multi	Act	Foot, Self, Pred	N
BeamDojo [28]	2025	RSS	B	Multi	Elev, Cont	Foot, Pred	N
DPL [53]	2025	RA-L	B	Ext, Multi	Lat	Unc, Pred	N
Gholami et al. [33]	2025	Sensors	B	Ext	Cont	Geo	N
HTCplanner [41]	2025	AI Enabled Robotic.	B	Ext	Cont, Vol	Geo, Plan	Y
Lin et al. [46]	2025	RA-L	Q	Pro, Multi	Sem	Prop	N
Medeiros et al. [80]	2025	RA-L	Q	Multi	Elev, Cont	Geo, Prop, Foot	N
Miguel et al. [30]	2025	IEEE Access	Q	Multi	Lat	Self	Y
PolygMap [42]	2025	ICRA	B	Multi	Cont, Sem	Geo, Sem, Plan	Y
Roth et al. [52]	2025	RSS	Q	Multi	Act	Pred	Y
Sanchez-Delgado et al. [48]	2025	iSpaRo	Q	Pro	Exp	Prop	N
Splatblox [73]	2025	arXiv	Q	Multi	Vol	Found	Y
STATE-NAV [21]	2025	RA-L	B	Ext	Elev, Exp, Act	Foot, Self, Unc, Pred	Y
Sun et al. [50]	2025	IROS	B	Multi	Elev, Sem	Prop, Pred	N
TANGO [58]	2025	ICRA	Q	Ext	Sem	Sem, Found	Y
Zhang et al. [31]	2025	Electronics	Q	Pro	Sem	Prop	N
Ægidius et al. [45]	2025	ICRA	Q	Ext	Cont, Sem, Lat	Self, Unc, Found	Y
CATNAV [62]	2026	arXiv	Q	Ext	Sem	Sem, Plan, Found	N
Compton et al. [29]	2026	arXiv	B	Ext, Multi	Elev	Pred	N
FocusNav [37]	2026	arXiv	B	Multi	Lat	Unc, Plan	N
GSAT [51]	2026	ICRA	Q	Ext	Lat	Geo, Unc	Y
Hindel et al. [81]	2026	arXiv	Q	Multi	Elev, Sem	Self, Sem, Prop	Y
TACT-ful [36]	2026	arXiv	B	Ext	Elev, Sem	Foot, Prop, Pred	Y
Xue et al. [24]	2026	RA-L	B	Ext	Vol	Pred	Y
Zhu et al. [54]	2026	arXiv	B	Ext, Multi	Cont, Lat	Foot, Pred	Y
CAT [77]	2026	arXiv	Q	Ext	Sem, Lat, Act	Sem, Self, Found	Y